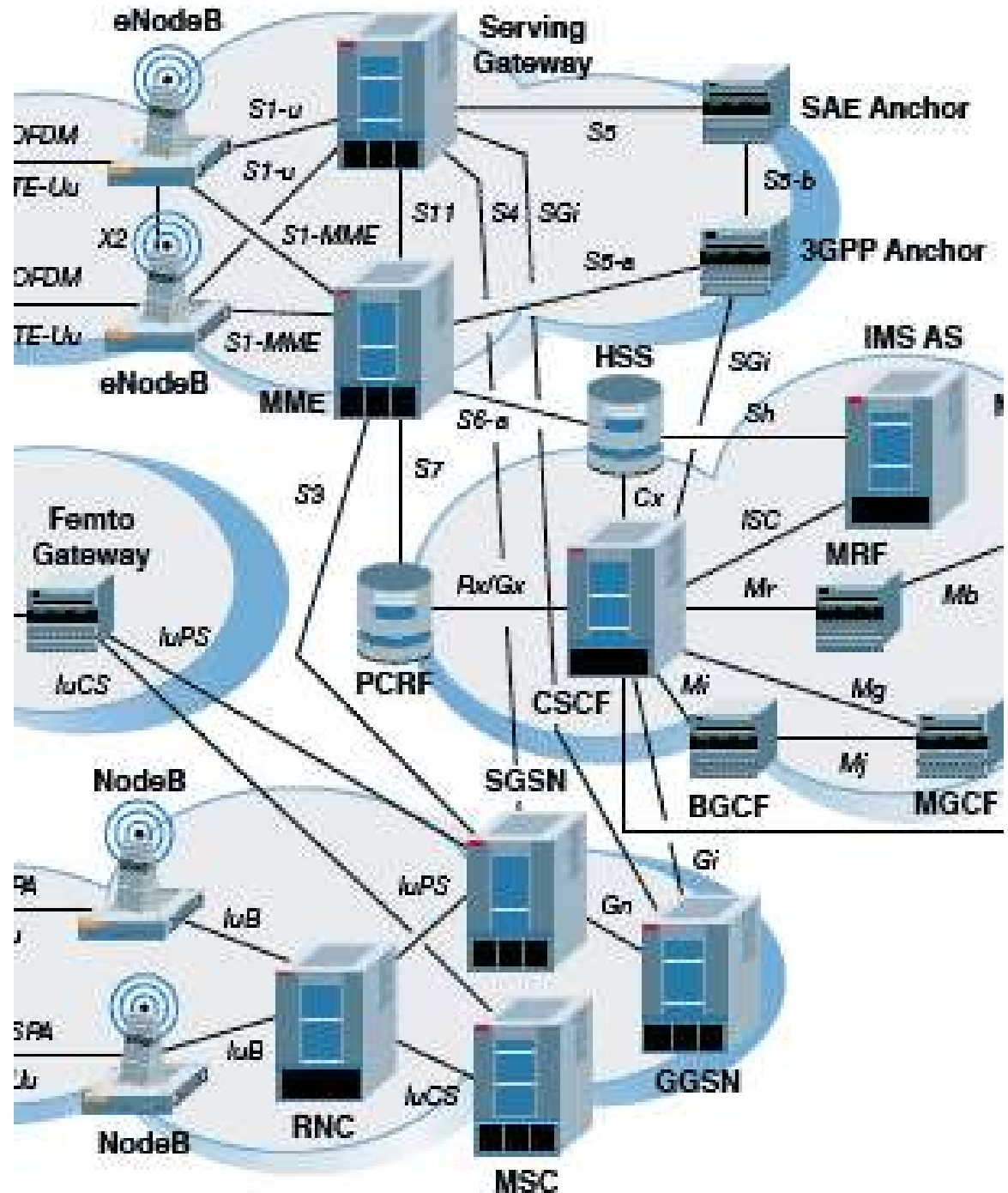


Chapter 3. Transmission & Switching Techniques

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12/22/2024



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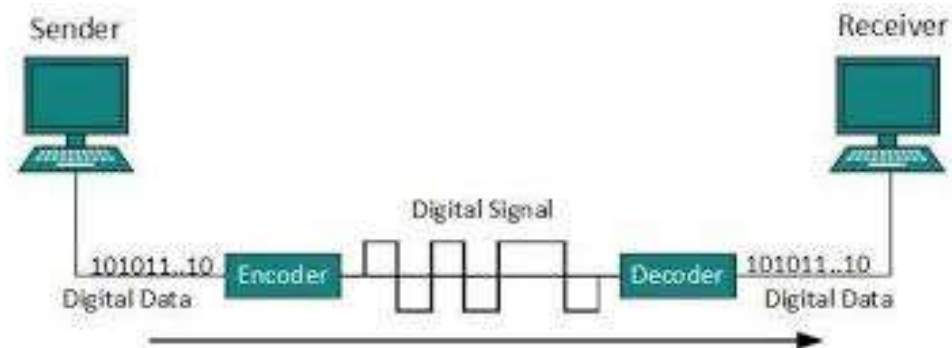
1. Definitions
2. Traffic flow
3. Transmission techniques
4. Switching techniques

1. Definitions

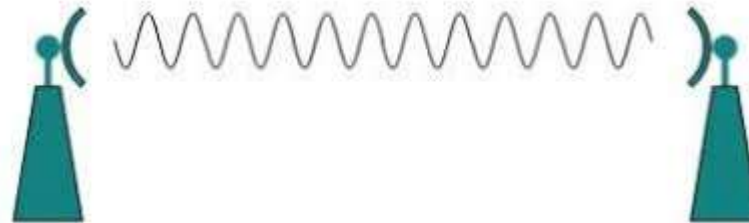
- **Transmission:** the **propagation** of a signal, message, or other form of intelligence **by any means** such as optical fiber, wire, or visual means.



Acoustic



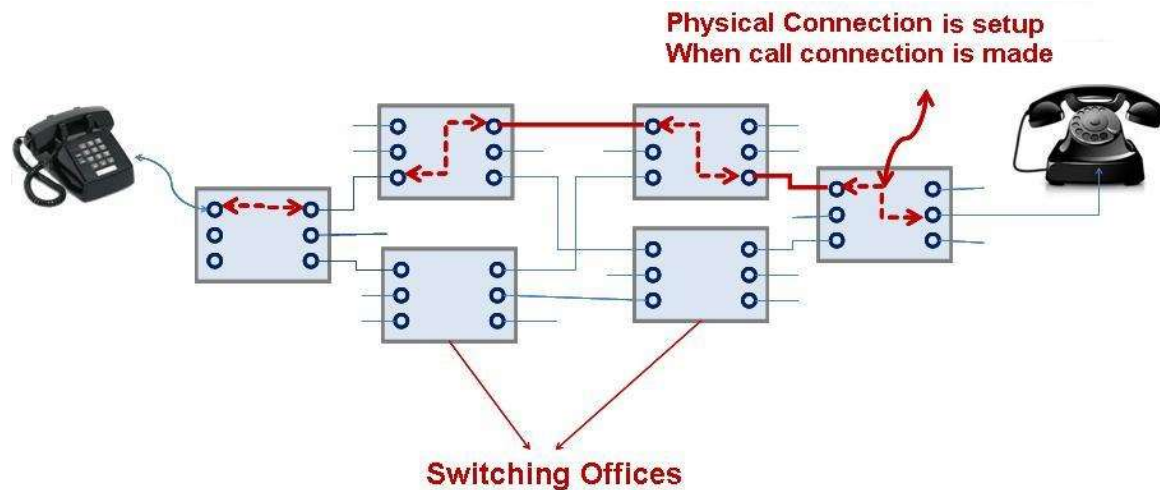
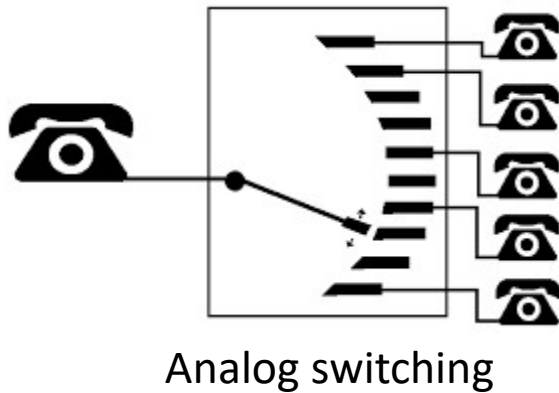
Wire



Wireless

1. Definitions (cont.)

- **Switching:** selects the route to the desired destination that the transmitted signal travels by the closing of switches in either the space domain, the time domain, frequency or any other forms.



Electronic circuit switching

2. Traffic flow

- Measurement of telephone traffic:

$$A = C \times T$$

A: Traffic flow (Erlang)

C: Number of calls originated during one hour

T: Average holding time (hours)

2. Traffic flow (cont.)

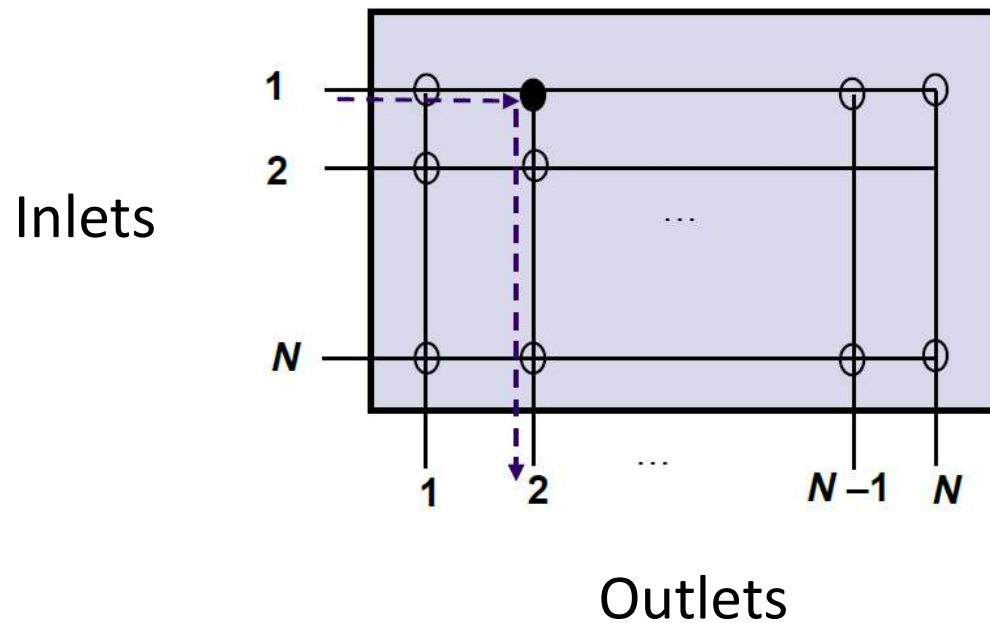
- Blockage, lost calls, and grade of service (GoS):
 - Example: tel. exchange serves 5000 subscribers, no more than 10% can connect simultaneously. When the number of served subscribers is 500, the subscriber 501th cannot connect. This call is a **lost call/ blocked** call and that subscriber has met **blockage**.
 - GoS express the **probability of meeting blockage** during **Busy hours (BH)**

2. Traffic flow (cont.)

- Example: determine the GoS if there are 354 lines connected for service and 6 blocked calls during BH.

2. Traffic flow (cont.)

- Availability: a switch has **full-availability** when each inlet has access to any outlet. Otherwise, **limited availability**



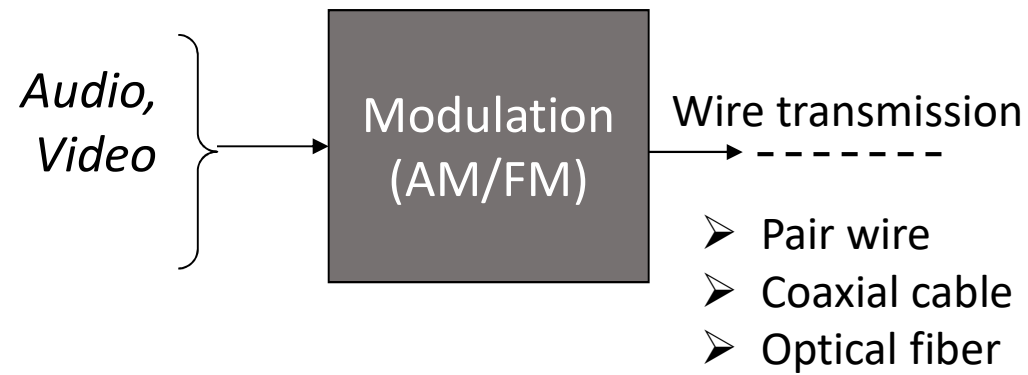
2. Traffic flow (cont.)

Table 4.1 Trunk-Loading Capacity, Based on Erlang B Formula, Full Availability

Trunks	Grade of Service 1 in 1000		Grade of Service 1 in 500		Grade of Service 1 in 200		Grade of Service 1 in 100		Grade of Service 1 in 50		Grade of Service 1 in 20	
	UC	TU	UC	TU	UC	TU	UC	TU	UC	TU	UC	TU
1	0.04	0.001	0.07	0.002	0.2	0.005	0.4	0.01	0.7	0.02	1.8	0.05
2	1.8	0.05	2.5	0.07	4	0.11	5.4	0.15	7.9	0.22	14	0.38
3	6.8	0.19	9	0.25	13	0.35	17	0.46	22	0.60	32	0.90
4	16	0.44	19	0.53	25	0.70	31	0.87	39	1.09	55	1.52
5	27	0.76	32	0.90	41	1.13	49	1.36	60	1.66	80	2.22
6	41	1.15	48	1.33	58	1.62	69	1.91	82	2.28	107	2.96
7	57	1.58	65	1.80	78	2.16	90	2.50	106	2.94	135	3.74
8	74	2.05	83	2.31	98	2.73	113	3.13	131	3.63	163	4.54
9	92	2.56	103	2.85	120	3.33	136	3.78	156	4.34	193	5.37
10	111	3.09	123	3.43	143	3.96	161	4.46	183	5.08	224	6.22
11	131	3.65	145	4.02	166	4.61	186	5.16	210	5.84	255	7.08
12	152	4.23	167	4.64	190	5.28	212	5.88	238	6.62	286	7.95
13	174	4.83	190	5.27	215	5.96	238	6.61	267	7.41	318	8.83
14	196	5.45	213	5.92	240	6.66	265	7.35	295	8.20	350	9.73
15	219	6.08	237	6.58	266	7.38	292	8.11	324	9.01	383	10.63
16	242	6.72	261	7.26	292	8.10	319	8.87	354	9.83	415	11.54
17	266	7.38	286	7.95	318	8.83	347	9.65	384	10.66	449	12.46
18	290	8.05	311	8.64	345	9.58	376	10.44	414	11.49	482	13.38
19	314	8.72	337	9.35	372	10.33	404	11.23	444	12.33	515	14.31
20	339	9.41	363	10.07	399	11.09	433	12.03	474	13.18	549	15.25
21	364	10.11	388	10.79	427	11.86	462	12.84	505	14.04	583	16.19
22	389	10.81	415	11.53	455	12.63	491	13.65	536	14.90	617	17.13
23	415	11.52	442	12.27	483	13.42	521	14.47	567	15.76	651	18.08
24	441	12.24	468	13.01	511	14.20	550	15.29	599	16.63	685	19.03
25	467	12.97	495	13.76	540	15.00	580	16.12	630	17.50	720	19.99
26	493	13.70	523	14.52	569	15.80	611	16.96	662	18.38	754	20.94
27	520	14.44	550	15.28	598	16.60	641	17.80	693	19.26	788	21.90

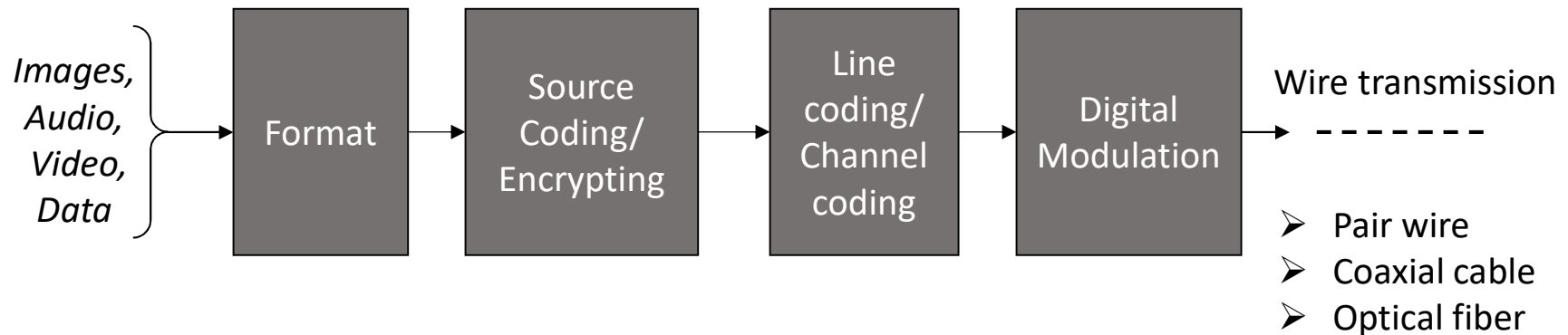
3. Transmission techniques

- Wire transmission for analog signal



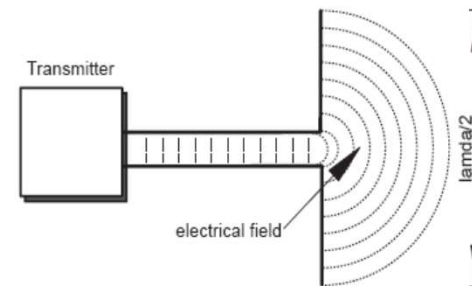
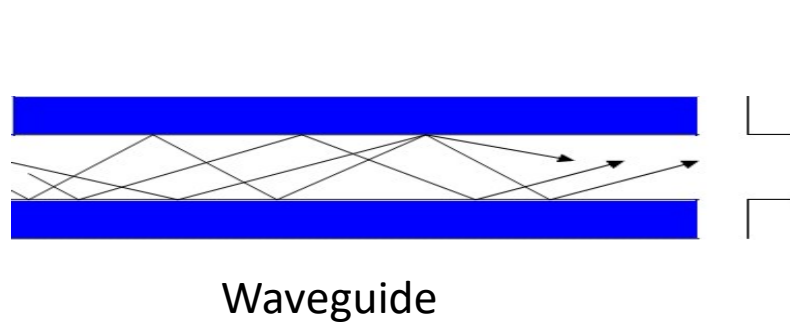
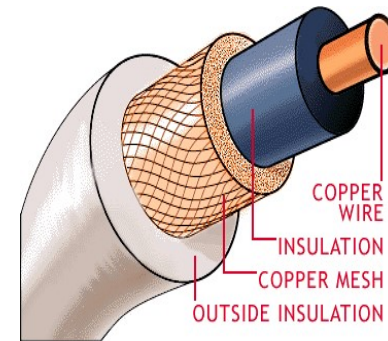
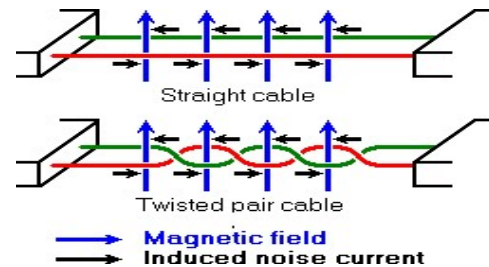
3. Transmission techniques(cont.)

- Wire transmission for digital signal



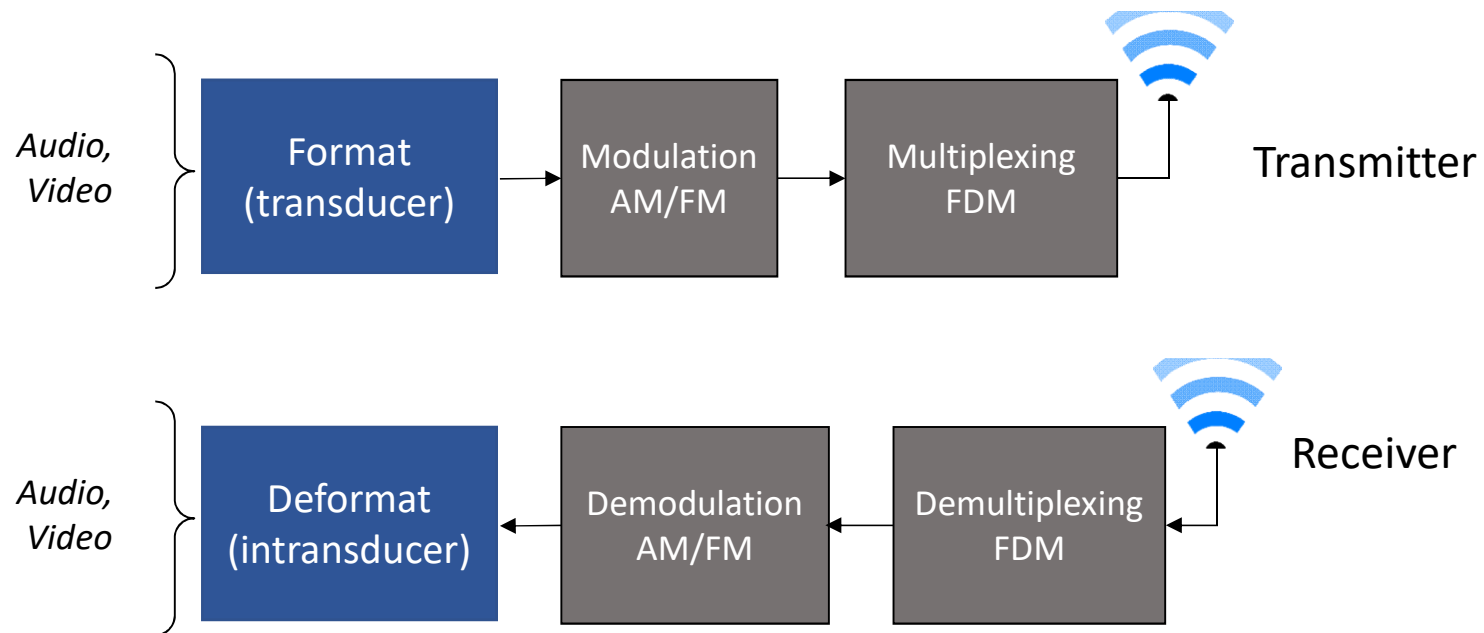
3. Transmission techniques(cont.)

- Wire transmission:



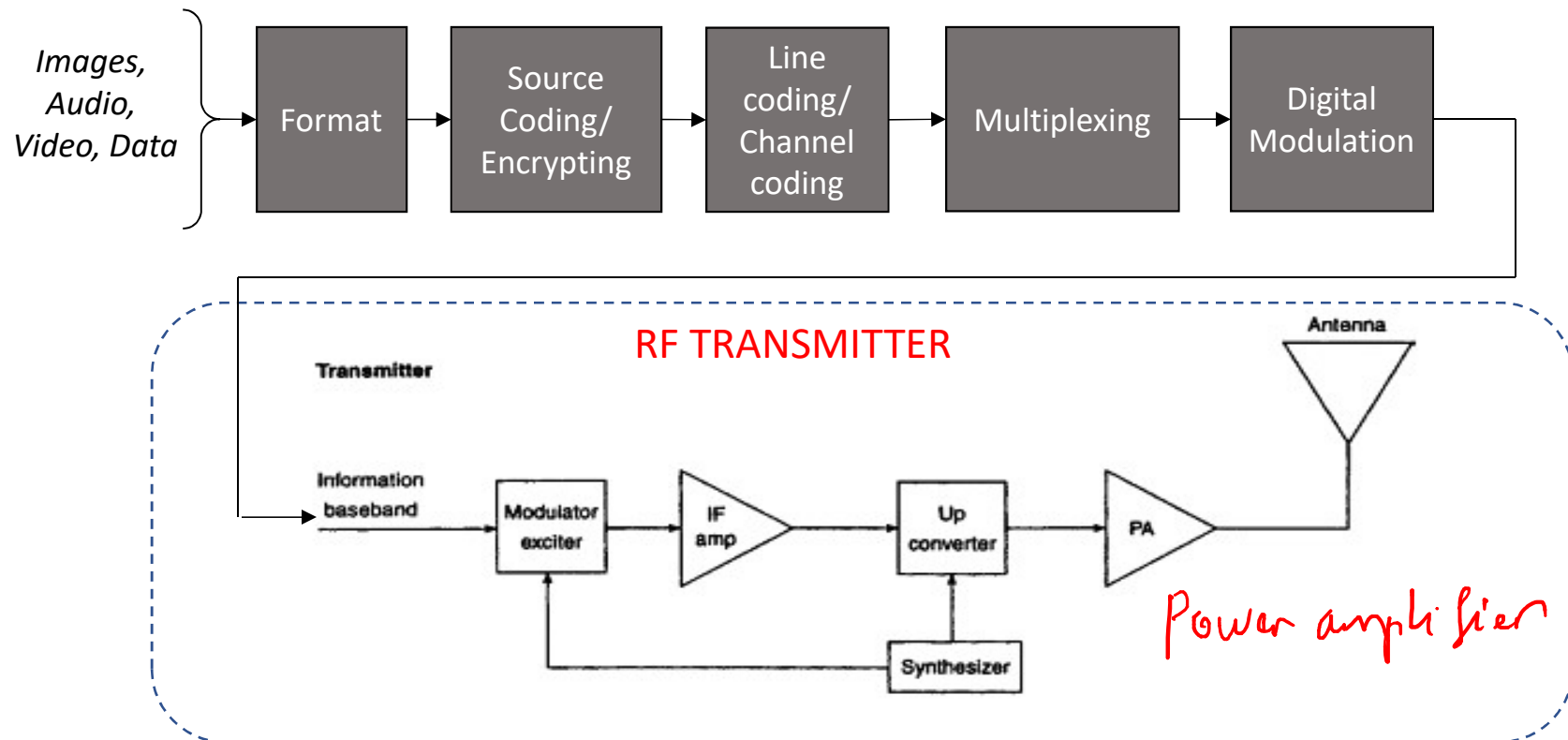
3. Transmission techniques(cont.)

- Wireless transmission for analog signal



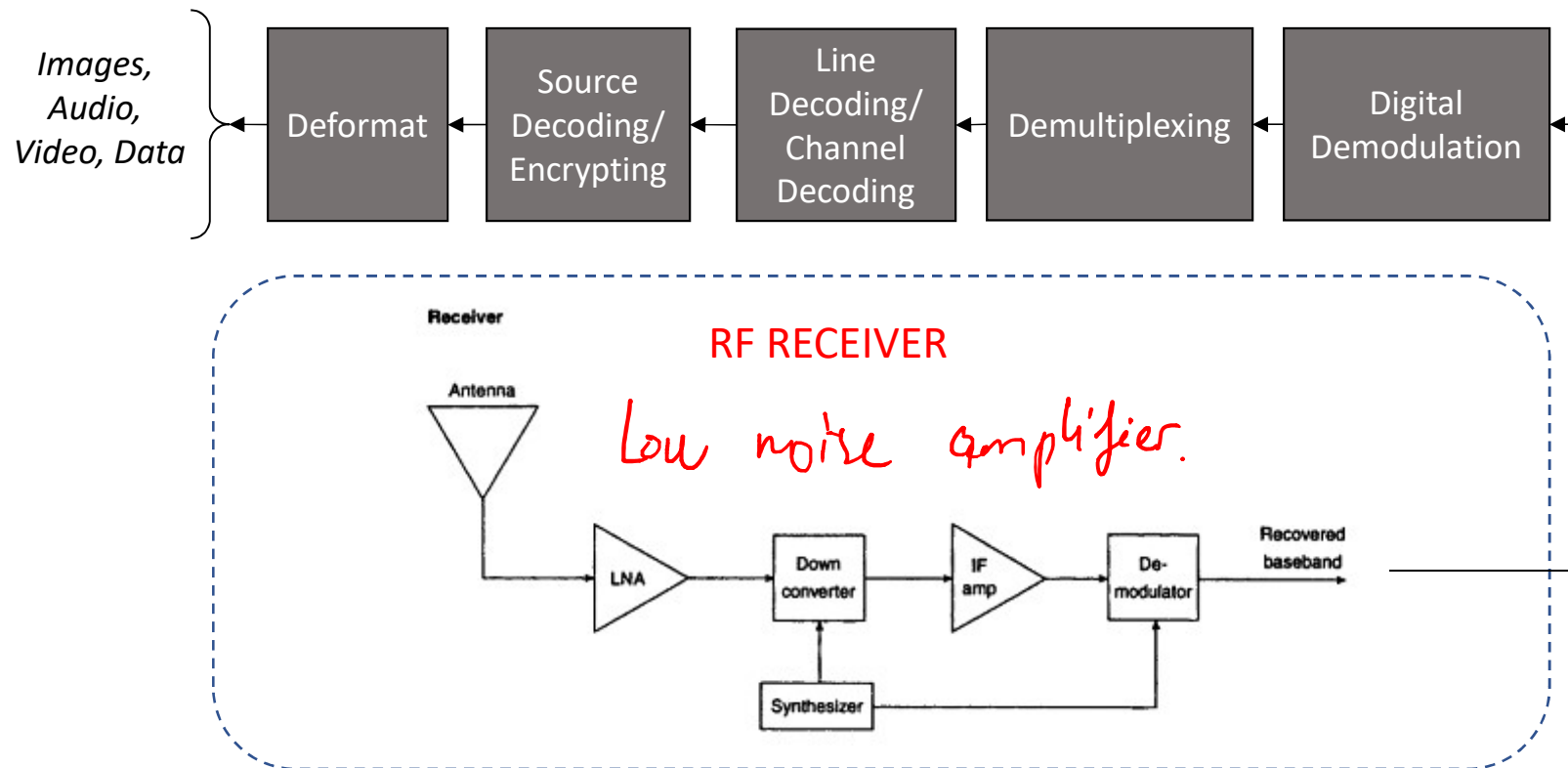
3. Transmission techniques(cont.)

- Wireless transmission for digital transmission



3. Transmission techniques (cont.)

- Wireless transmission for digital transmission



3. Transmission techniques(cont.)

- Why do we modulate many stages for radio transmission?

Comparison

Analog transmission	Digital transmission
Advantage	Disadvantage
Disadvantage	Advantage

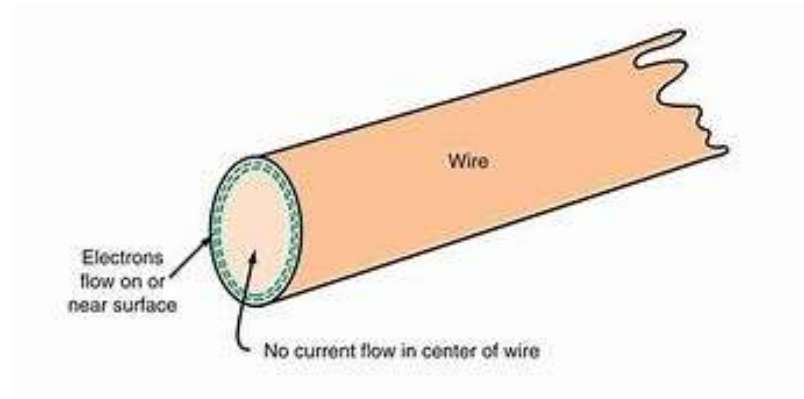
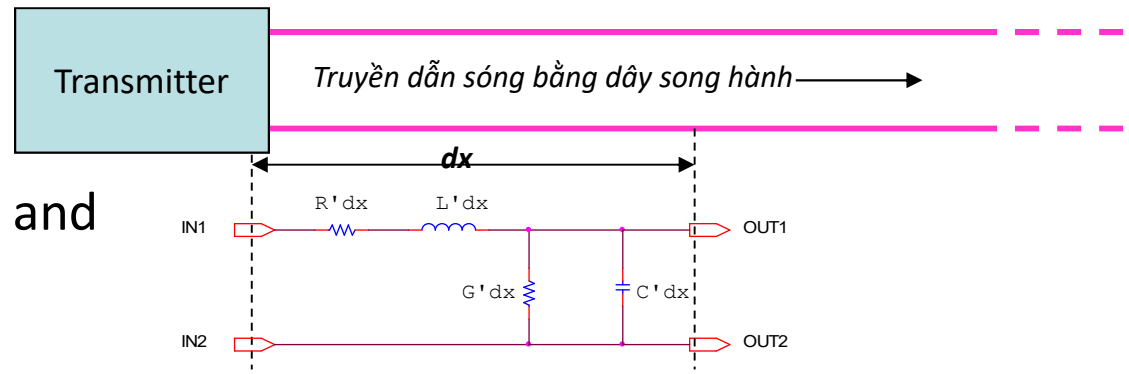
3.1. Wire transmission

- General characteristics: attenuation (loss), degradation/distortion, delay and noise

$$L_{dB} = 10 \log \frac{P_{in}}{P_{out}} \quad (\text{dB})$$

$$PSNR = 20 \log \left(\frac{\text{Peak_signal}}{\text{Noise}} \right)$$

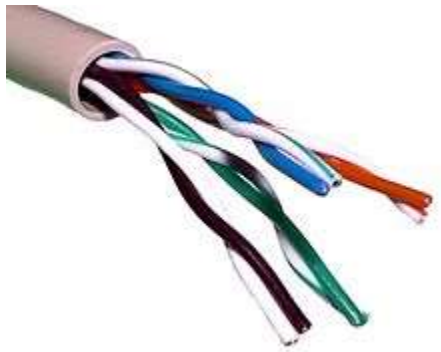
Skin effect creates a change of the characteristic resistance of the transmission lines



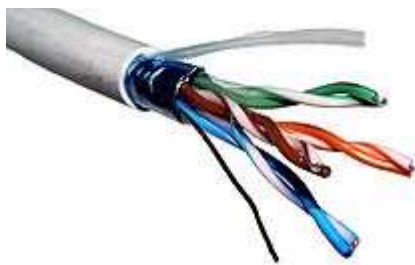
3.1. Wire transmission (cont.)

- Ex: Suppose a 10-mW (milli watt), 1000-Hz signal is launched into a wire pair. At the distant end of the wire pair the signal is measured at 0.2 mW. What is the loss in decibels on the line for this signal?

Twisted pair cables

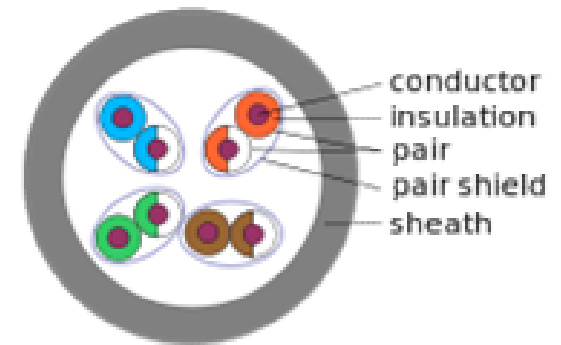


Unshielded twisted pair (UTP)

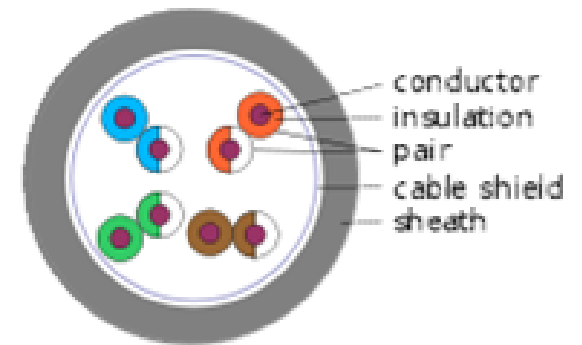


Shielded twisted pair (STP)

STP



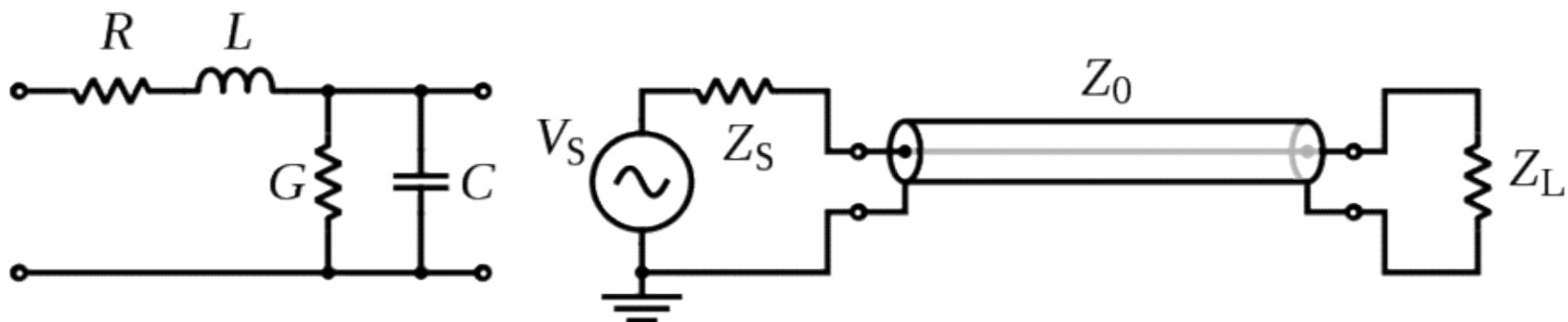
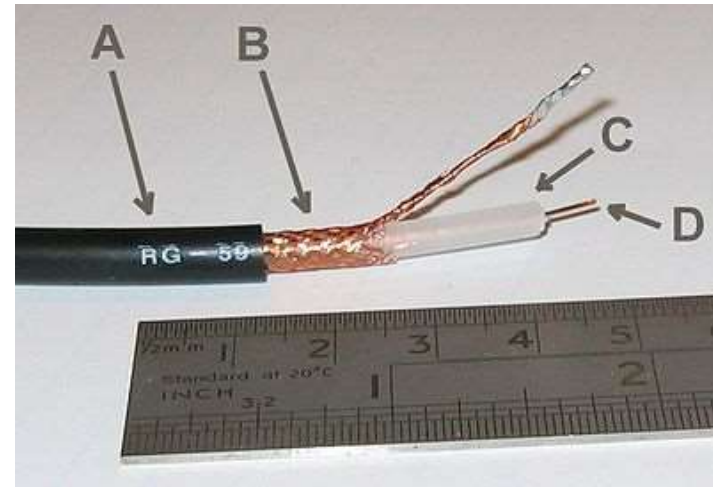
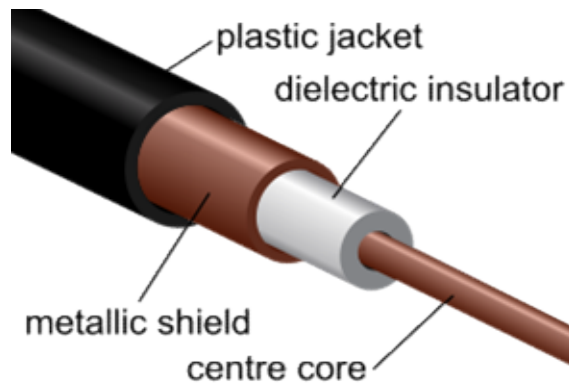
S/UTP



Twisted pairs characteristics

16MHz ^[6]	10BASE-T and 100BASE-T4 Ethernet ^[6]	Described in EIA/TIA-568. Unsuitable for speeds above 16 Mbit/s. Now mainly for telephone cables ^[6]	
UTP ^[6]	20MHz ^[6]	16 Mbit/s ^[6] Token Ring	Not commonly used ^[6]
UTP ^[6]	100MHz ^[6]	100BASE-TX & 1000BASE-T Ethernet ^[6]	Common in most current LANs ^[6]
UTP ^[6]	100MHz ^[6]	100BASE-TX & 1000BASE-T Ethernet ^[6]	Enhanced Cat5. Same construction as Cat5, but with better testing standards.
UTP ^[6]	250MHz ^[6]	1000BASE-T Ethernet	Most commonly installed cable in Finland according to the 2002 standard. SFS-EN 50173-1
	250MHz (500MHz according to some)	Not a standard; a cable maker's own label.	
	500MHz	10GBASE-T Ethernet	ISO/IEC 11801:2002 Amendment 2.
S/FTP ^[6]	600MHz ^[6]	Telephone, CCTV , 1000BASE-TX in the same cable. 10GBASE-T Ethernet .	Four pairs, U/FTP (shielded pairs). Standard under development.
	1000MHz	Telephone, CATV , 1000BASE-TX in the same cable. 10GBASE-T Ethernet .	Four pairs, S/FTP (shielded pairs, braid-screened cable). Standard under development.

Coaxial Cables

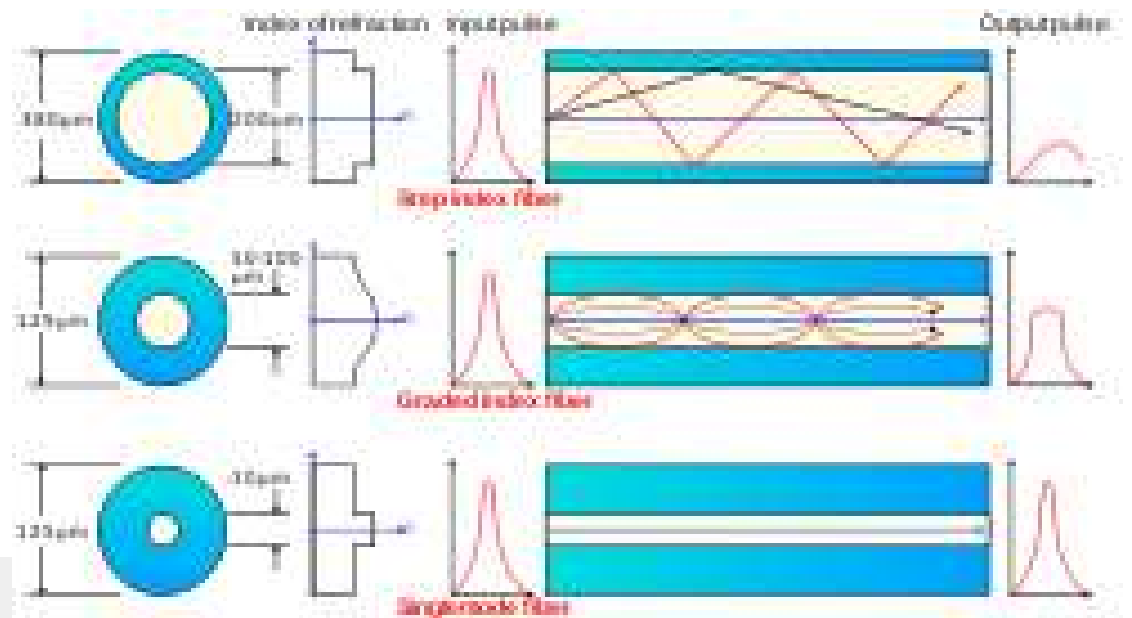


Range up to 10 GHz for the transmission frequency

Fiber-Optic cables



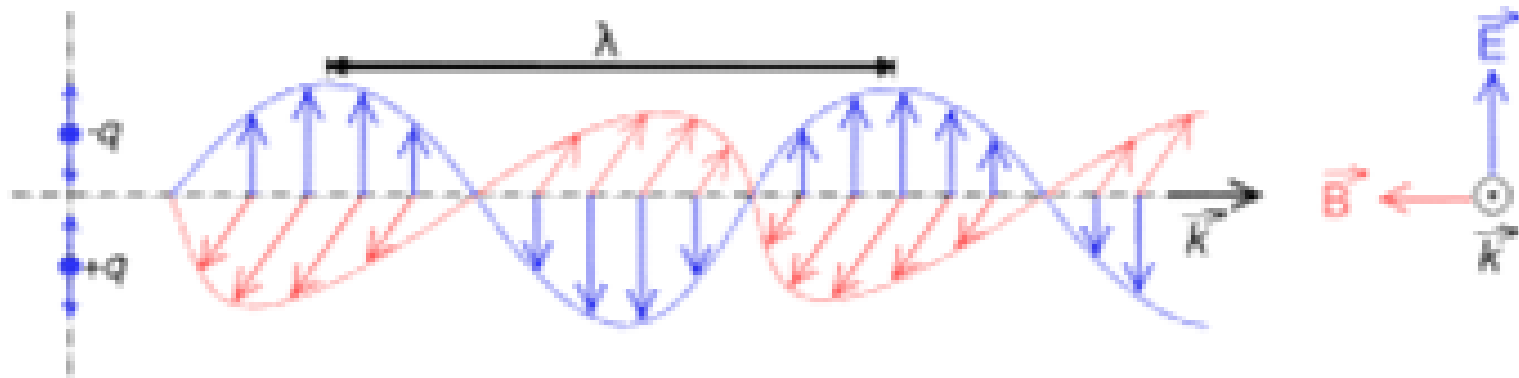
Transfer mode in fiber-optic cables



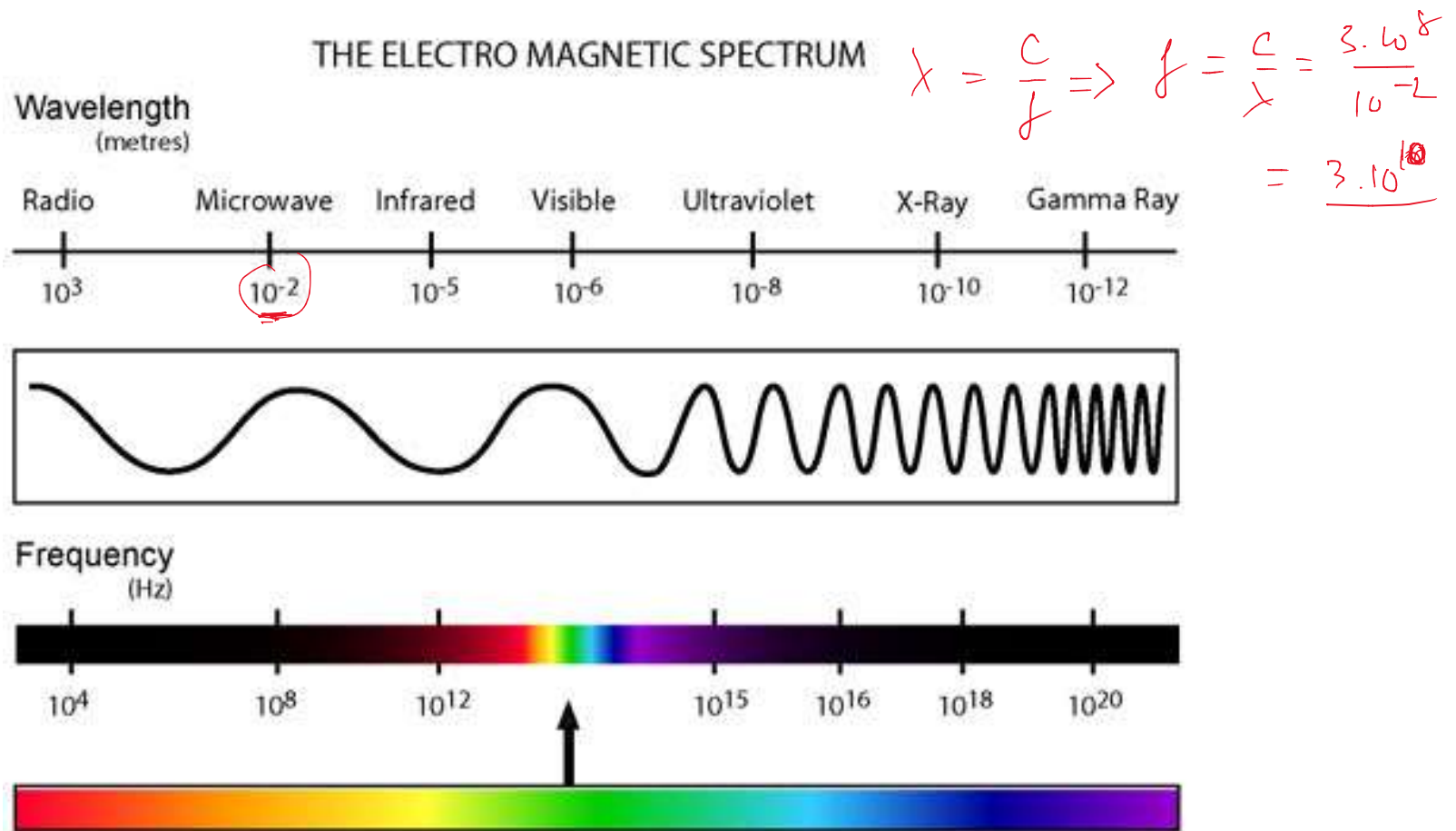
Range up to 100 Petabit per second

3.2. Wireless (Radio) transmission

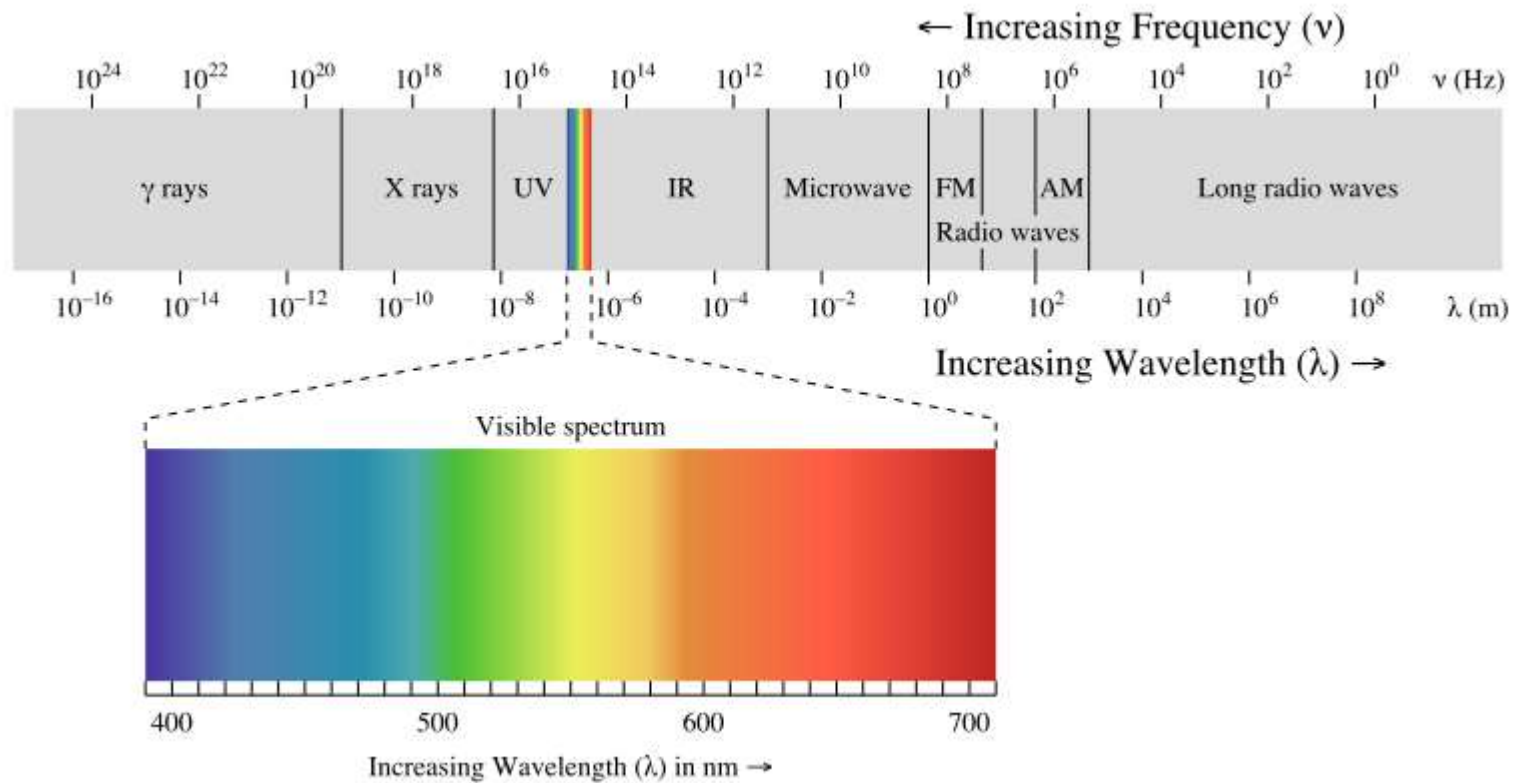
Electromagnetic Waves



3.2. Wireless (Radio) transmission

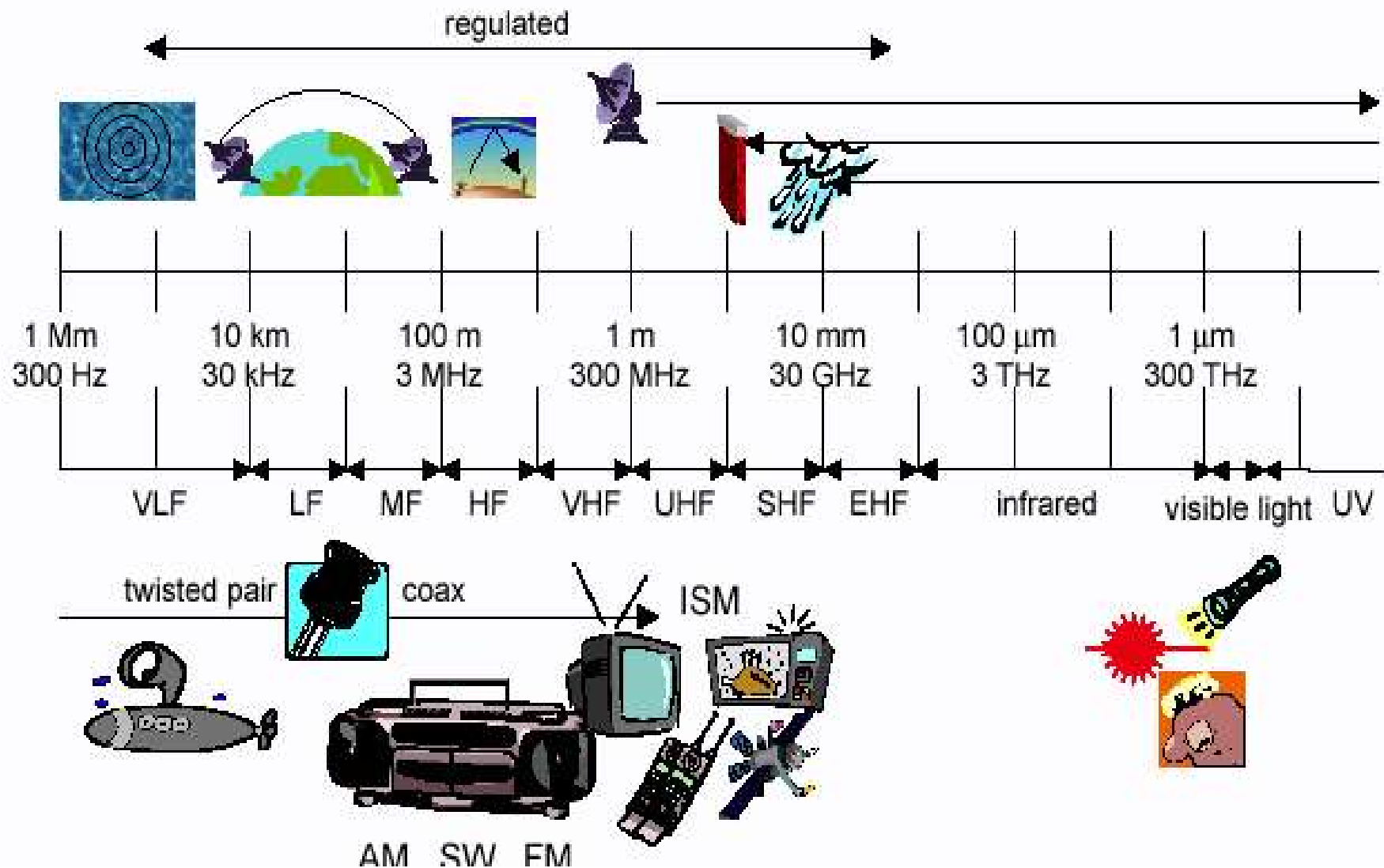


Electromagnetic Spectrum



http://www.edumedia-sciences.com/a185_l2-transverse-electromagnetic-wave.html

Electromagnetic Spectrum



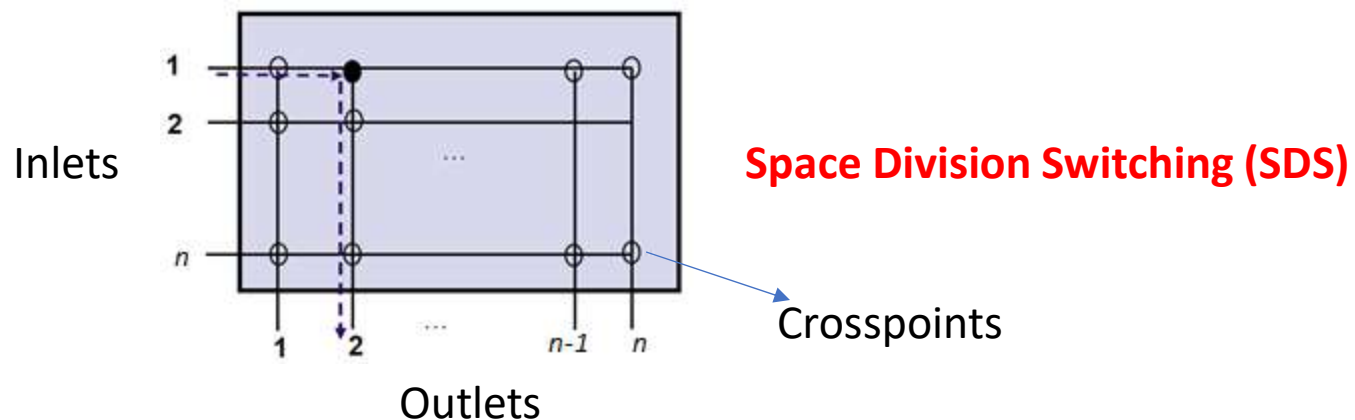
Wave length	Frequency Designations	Transmission Media	Propagation Modes	Representative Applications	Frequency
1 cm	Extra High Frequency (EHF)	Wave guide		Satellite, Microwave relay, Earth-satellite radar.	100 GHz
10 cm	Super High Frequency (SHF)				10 GHz
1 m	Ultra High Frequency (UHF)	Coaxial Cable	Line-of-sight radio	Wireless comm. service, Cellular, pagers, UHF TV	1 GHz
10m	Very High Frequency (VHF)			Mobile, Aeronautical, VHF TV and FM, mobile radio	100 MHz
100m	High Frequency (HF)			Amateur radio, Civil Defense	10 MHz
1 km	Medium High Frequency (MF)	Wire pairs	Ground wave radio	AM broadcasting	1 MHz
10 km	Low Frequency (LF)			Aeronautical, Submarine cable, Navigation,	100 kHz
100km	Very Low Frequency (VLF)			Transoceanic radio	10 kHz

4. Switching techniques

- **Switching** connects and transmits a signal between the source and the destination.
- Three typical switching for digital traffic:
 - Circuit Switching
 - Message Switching
 - Packet Switching

4.1 Circuit switching

- **Circuit switching** is a technique that directly connects the sender and the receiver in an unbroken path.
- With this type of switching technique, once a connection is established, a dedicated path exists between both ends until the connection is terminated.
- Routing decisions must be made when the circuit is first established, but there are no decisions made after that time

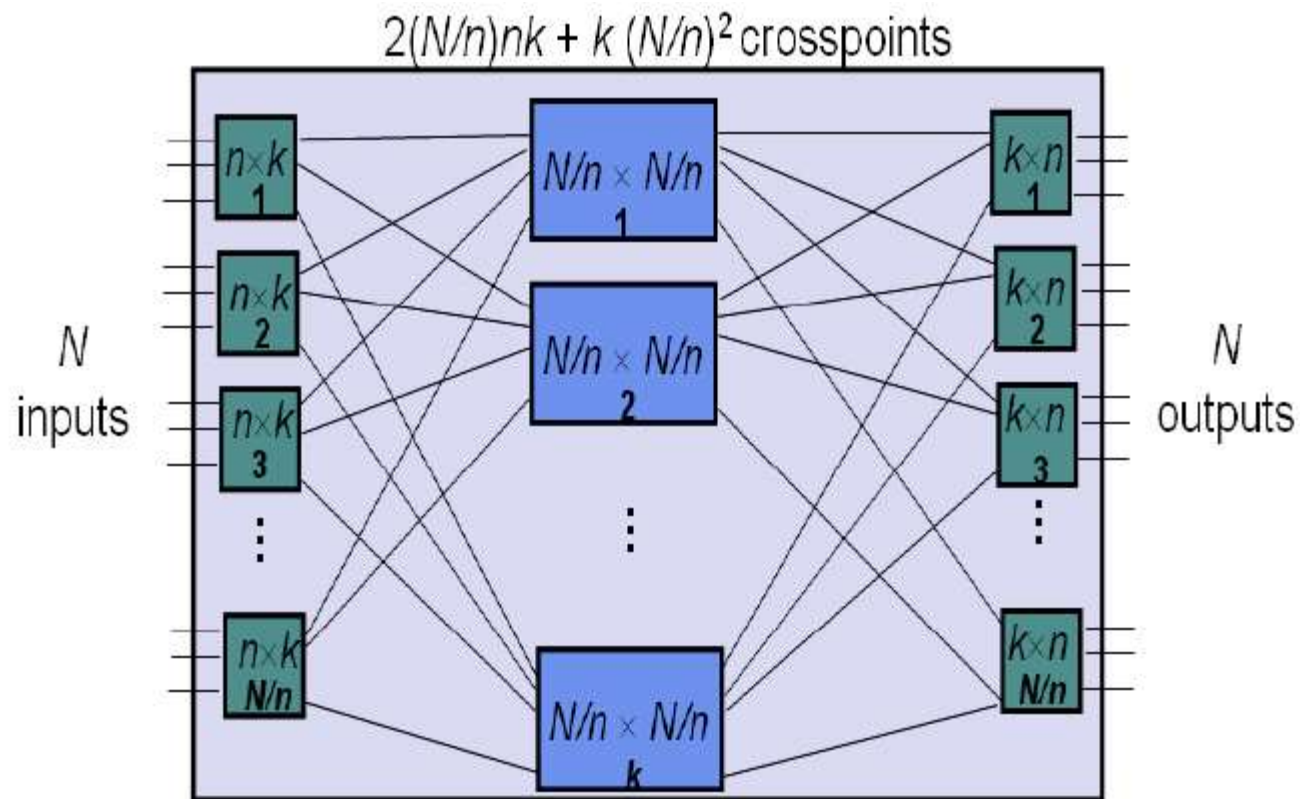


4.1 Circuit switching (cont.)

- **Circuit switching** in a network operates almost the same way as the telephone system works.
- A complete **end-to-end path** must exist before communication can take place.
- The computer initiating the data transfer must ask for a connection to the destination.
- Once the connection has been initiated and completed to the destination device, the destination device must acknowledge that it is ready and willing to carry on a transfer.

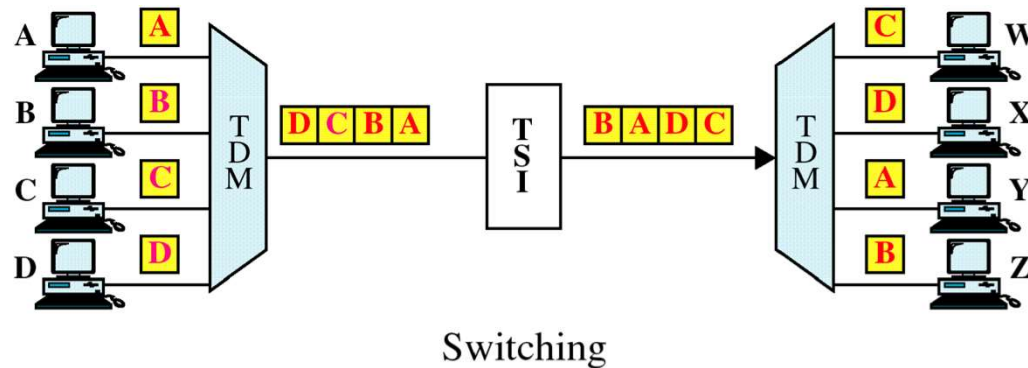
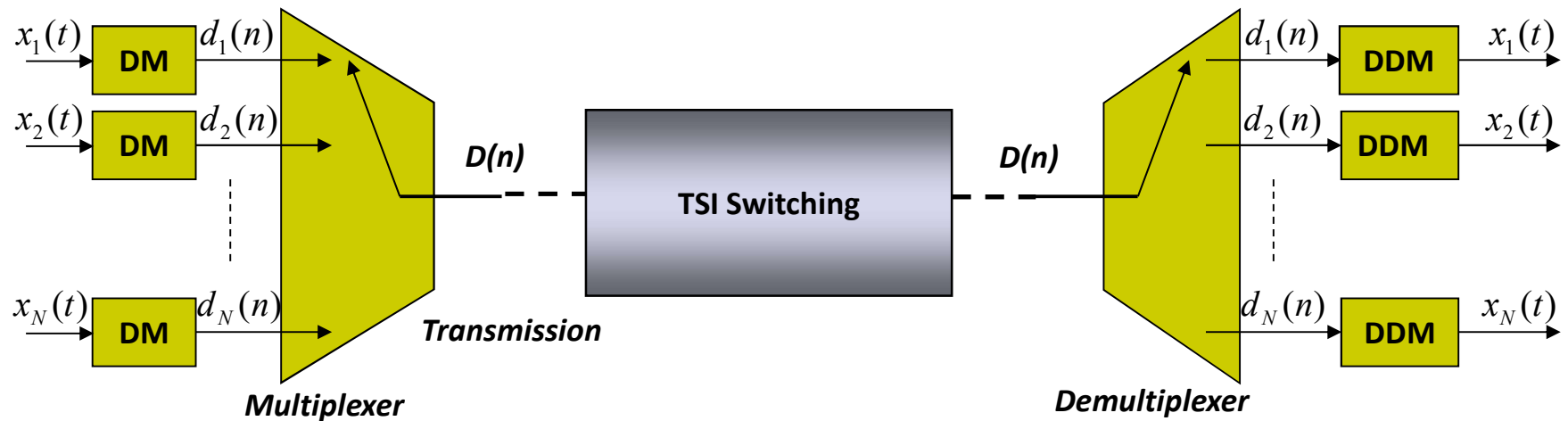
4.1 Circuit switching (cont.)

Space division switching with multi-stages



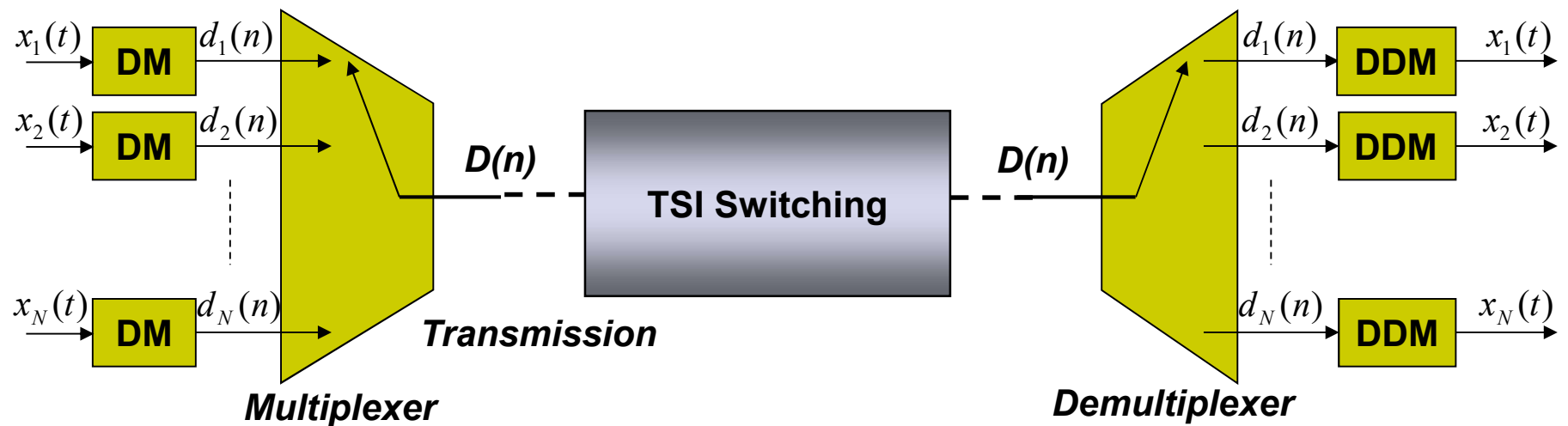
4.1 Circuit switching (cont.)

Time slot interchange (TSI) or Time division multiplex -TDM



4.1 Circuit switching (cont.)

Examples,



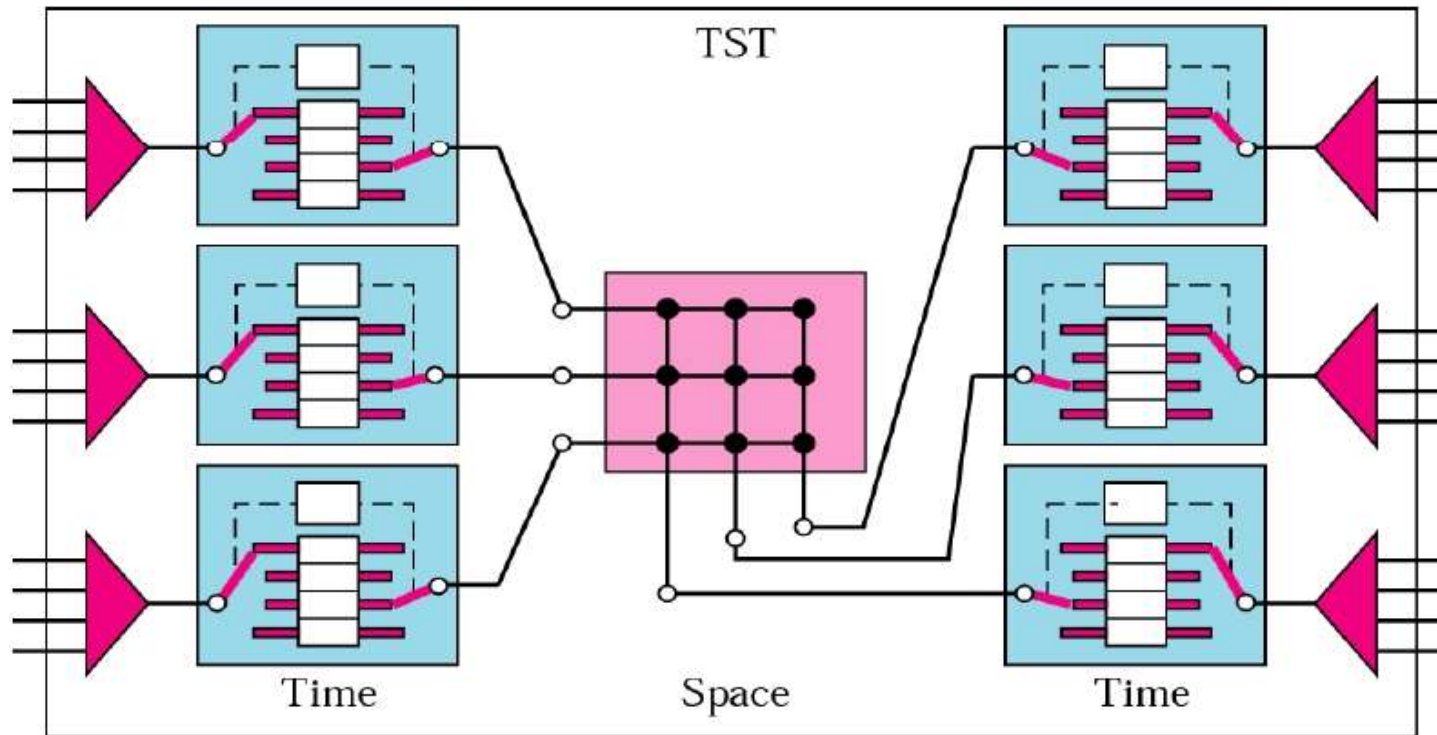
- Sampled frequency: 8KHz
- Switching frequency (or the clock of TSI): 200MHz
 - Define the rate of $d_n(n)$
 - Define the maximum capacity of the TSI

Comparison

SDS	TSI/TDM
Advantage	Disadvantage
- Simple	- Complex
- Low cost	- Expensive
- High rate/ Larger BW	- Low rate/ Smaller BW
Disadvantage	Advantage
- High noise/ Low quality	- Low noise/ High quality
- Not ability for extend inlets/ cross points	- Ability for extend inlets/ cross points
- More increase cost when increase in large amount of cross points	- Not much increase cost when increase in large amount of cross points

4.1 Circuit switching (cont.)

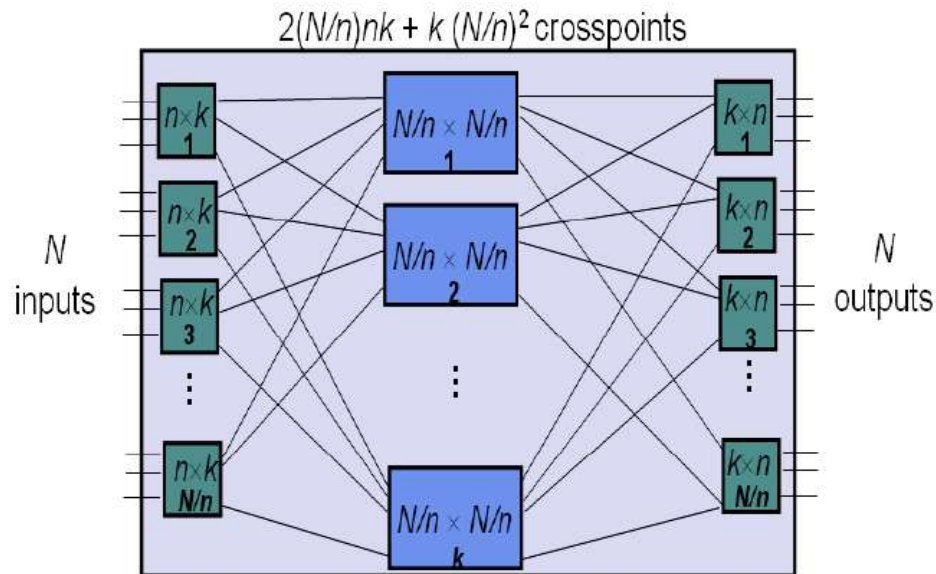
TST (Time-Space-Time)



4.1 Circuit switching (cont.)

- One attractive feature of a single stage switch is that it is strictly Nonblocking
- If the called party is idle, the desired connection can always be established by selecting the particular crosspoint dedicated to the particular input/output pair
- When crosspoints are shared, the possibility of blocking arises
- In 1953, Charles Clos of Bell Lab demonstrated that if each individual array is nonblocking, and if the number of center stages k is equal to $2n-1$, the switch is strictly nonblocking

Nonblocking switches for SDS



- **Nonblocking condition:**

- Connection through the three-stage switch requires locating a center-stage array with an idle link from the appropriate first stage and idle link to the appropriate third stage
- Individual arrays themselves are nonblocking
- Desired path can be set up any time a center stage with the appropriate idle links can be located

Nonblocking switches for SDS (cont.)

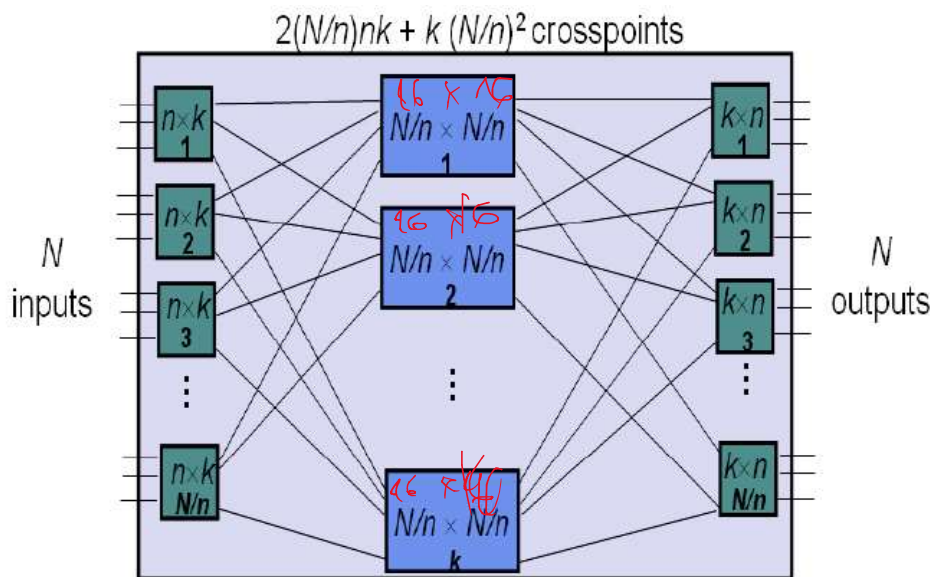
Three-stage switch is strictly nonblocking if

$$k = 2n - 1$$

- Optimum value of n , that yields the minimum number of crosspoints,

$$n_{opt} = \sqrt{\frac{N}{2}}$$

$$N_x(\min) = 4N(2n - 1)$$

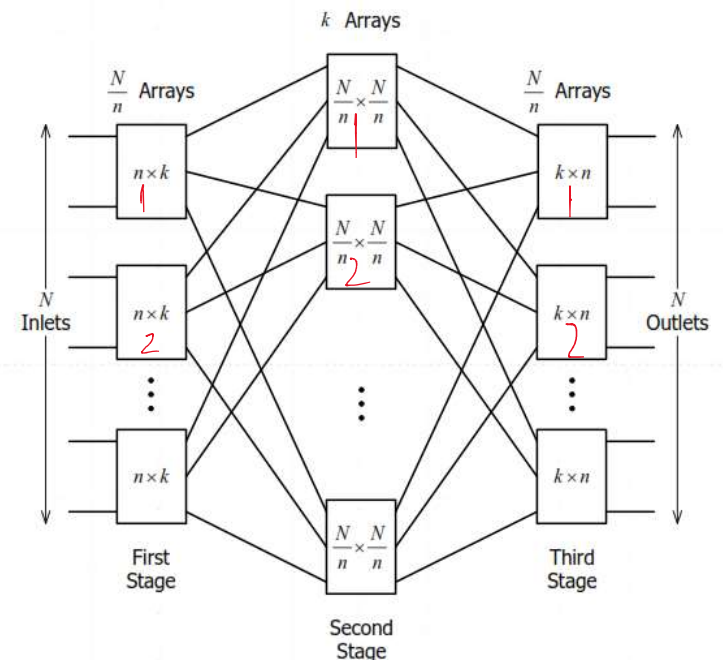


Blocking probabilities

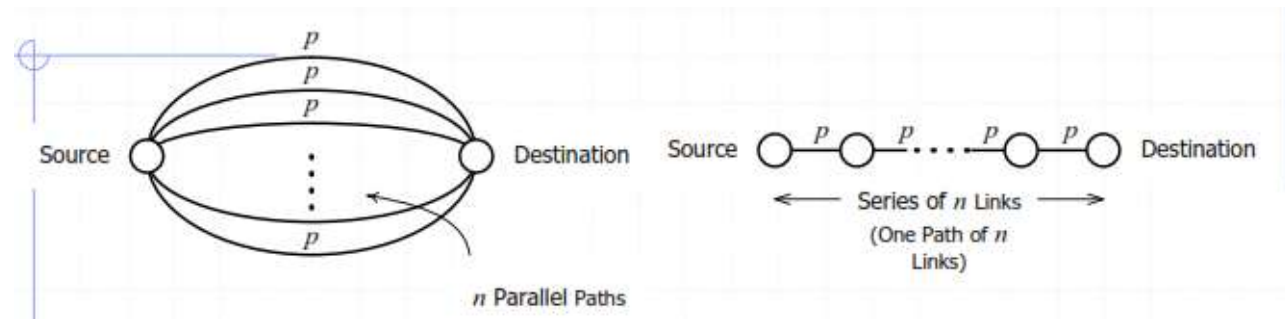
- Strictly nonblocking switches are rarely needed in most voice telephone networks
- Public telephone network is designed to provide a certain maximum probability of blocking for the busiest hour of the day
- Value of blocking probability is one aspect of the telephone company's grade of service
 - ✓ Availability
 - ✓ Transmission quality
 - ✓ Delay in setting up a call
- Typical residential telephone is busy 5-10% of the time during the busy hour
- Business telephone is often busy for a larger percentage of their busy hour
- Network-blocking occurrences on the order of 1% during the busy hour do not represent a significant reduction in the ability to communicate since the called party is much more likely to have been busy anyway

Blocking probabilities

- Calculating blocking probabilities proposed by C. Y. Lee
- Determining the blocking probabilities of various switching structures uses utilization percentage or “loading” of individual links
 - Notation p is used, in general, to represent the fraction of time that a particular link is in use (p is the probability that a link is busy)
 - p is sometimes referred to as an occupancy
 - Probability that a link is idle is denoted by $q = 1 - p$



Blocking probabilities



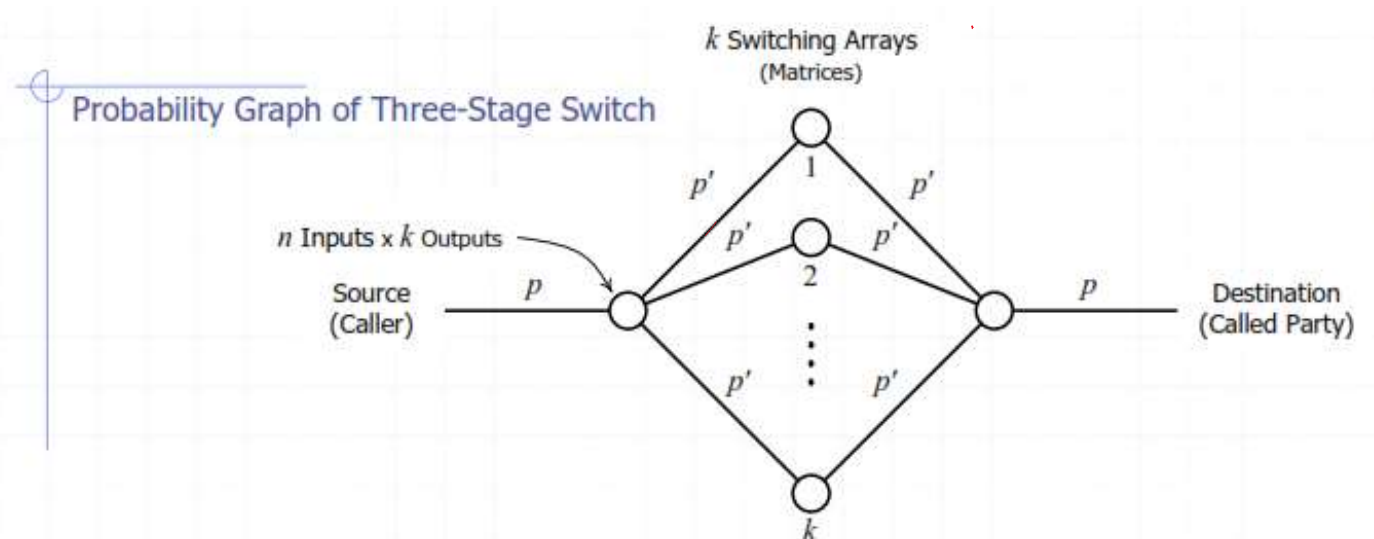
- When any one of n parallel paths can be used to complete a connection, the composite blocking probability B is the probability that all paths are busy
 - Each path is busy or idle independently of other paths

$$B = p^n$$

When a series of n links are all needed to complete a connection, the blocking probability is mostly easily determined as 1 minus the probability that they are all available. Each link is busy or idle independently of other links.

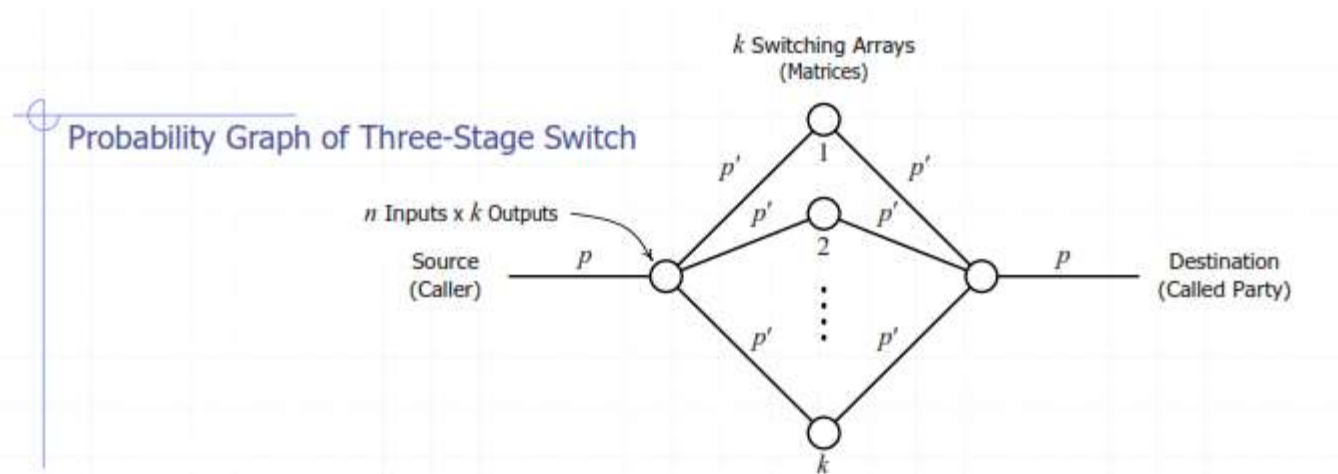
$$B = 1 - q^n = 1 - (1 - p)^n$$

Blocking probabilities



- ✓ Any particular connection can be established with k different paths
- ✓ One through each center-stage array
- ✓ Probability that an inlet is busy is denoted by p
- ✓ Probability that any particular interstage link is busy is denoted by p'
- ✓ The number of center-stage (second-stage) arrays is denoted by k

Blocking probabilities



- The probability of blocking for a three-stage switching can be determined as
B= prob. that all paths are busy
= (prob. that an arbitrary path is busy)^k
=(1-prob. that all the links in that path are available)^k

$$B = (1 - q'^2)^k = \left[1 - (1 - p')^2 \right]^k$$

Blocking probabilities

- Each first-stage array (switch) is of the size $n \times k$
- An inlet can use any of k outputs
- k outputs are shared by all n inputs
- If the probability p that an inlet is busy is known, the probability p' that inter-stage link is busy can be determined as

$$p' = p \cdot \frac{n}{k} = p / \beta \quad \text{với} \quad \beta = k / n$$

- That is, the percentage of inter-stage links that are busy is reduced by a factor of β
- The factor β is defined as though k is greater than n , which implies that the first stage of the switch is providing space expansion (i.e., switching some number of input links to a larger number of output links)

Blocking probabilities

- Actually, β may be less than 1, implying that the first stage is concentrating the incoming traffic
- First-stage concentration is usually employed in end-office or large PBX switches where inlets are highly used (5-10%)
- In tandem or toll offices, however, the incoming trunks are heavily utilized, and expansion is usually needed to provide adequately low-blocking probabilities
- the blocking probability of a three-stage switch in terms of the inlet utilization p is given by

$$B = \left[1 - (1 - p / \beta)^2 \right]^k$$

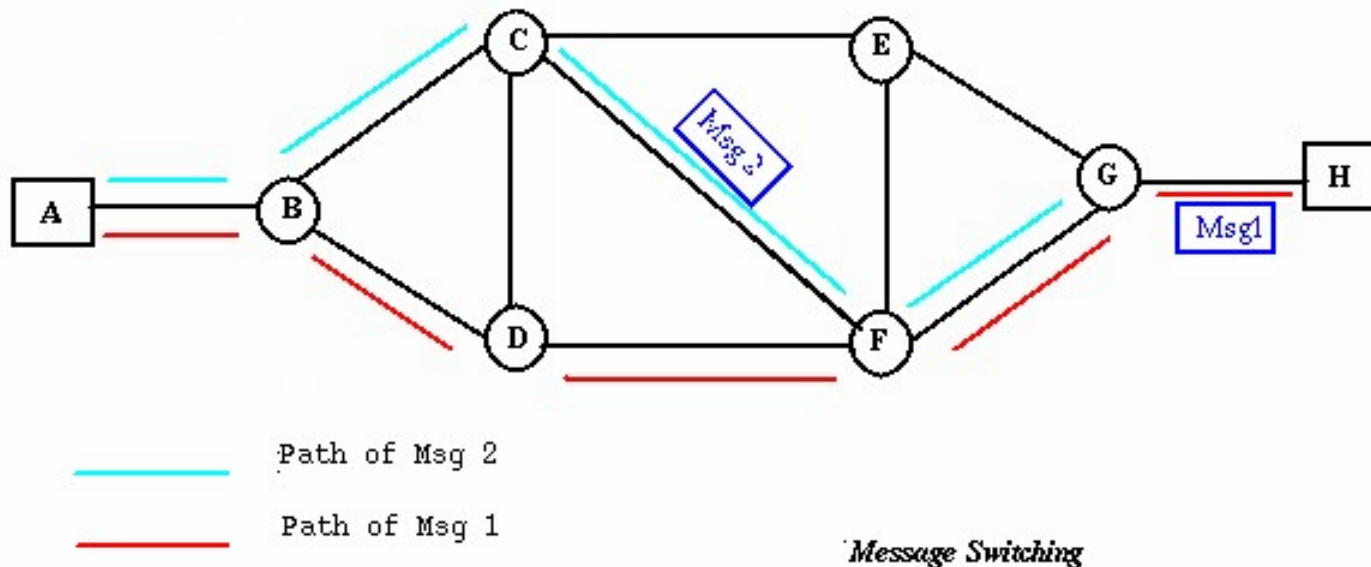
Exercise

Design a switch of a PABX (Private Automatic Branch Exchange). It is assumed that the PABX serves for 512 subscribers with blocking probability at 0.01 and utilization probability is 0.2 at the busy hour.

Design a SDS switch of a PABX (Private Automatic Branch Exchange). It is assumed that the PABX serves for 125 subscribers with blocking probability at 0.02 and utilization probability is 0.12 at the busy hour.

Design a switch of a PABX (Private Automatic Branch Exchange). It is assumed that the PABX serves for 512 subscribers with blocking probability at 0.01 and utilization probability is 0.1 at the busy hour.

4.2. Message Switching



- ✓ Signal is stored in a message and transmitted in many routs to the destination.
- ✓ The size of messages can be different and the time to receive the message can be long

4.2. Message Switching (cont.)

Characteristics:

- With message switching there is no need to establish a dedicated path between two stations.
- When a station sends a message, the destination address is appended to the message.
- The message is then transmitted through the network, in its entirety, from node to node.
- Each node receives the entire message, stores it in its entirety on disk, and then transmits the message to the next node.
- This type of network is called a store-and-forward network.

4.2. Message Switching (cont.)

- **Advantage:**

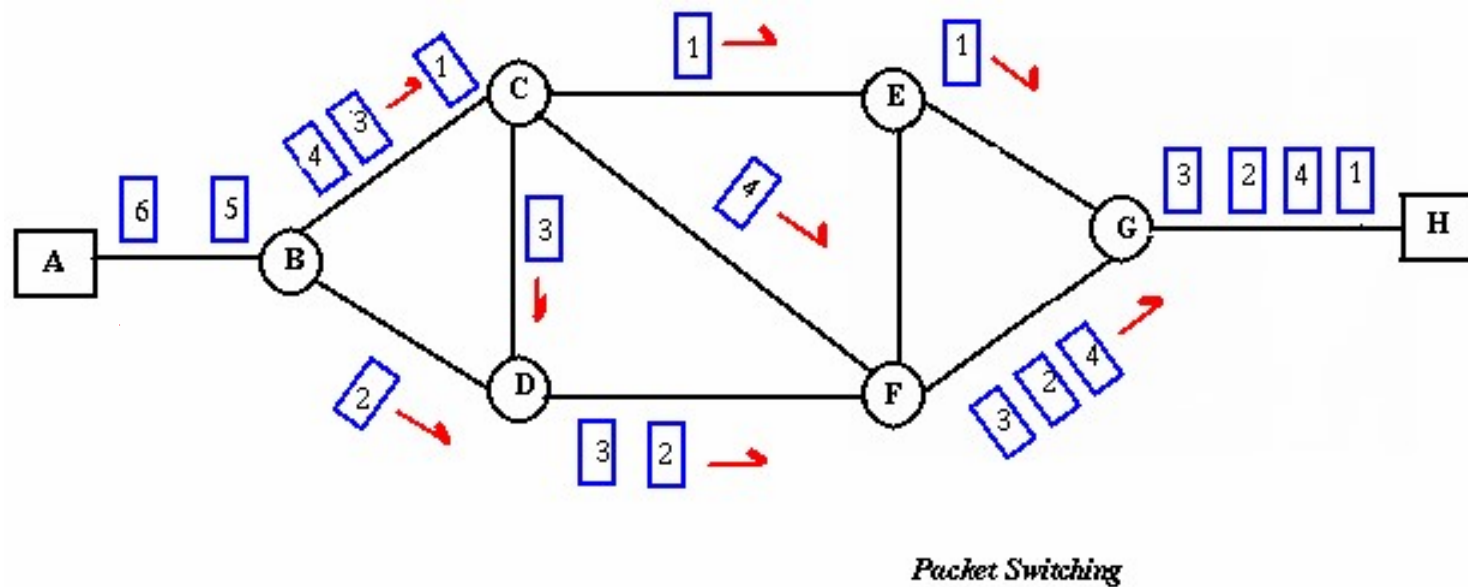
- Channel efficiency can be greater compared to circuit-switched systems, because more devices are sharing the channel.
- Traffic congestion can be reduced, because messages may be temporarily stored in route.
- Message priorities can be established due to store-and-forward technique.
- Message broadcasting can be achieved with the use of broadcast address appended in the message

4.2. Message Switching (cont.)

Disadvantage:

- The switching time is long and uncontrollable
- Message switching is not compatible with interactive applications.
- Store-and-forward devices are expensive, because they must have large disks to hold potentially long messages.

4.3. Packet Switching



A message is broken into small parts, called packets, with the same size. Each packet adds header, which include index, addresses of the source and the destination, error control code. Then the aligned data packets are sent to the destination by different routs.

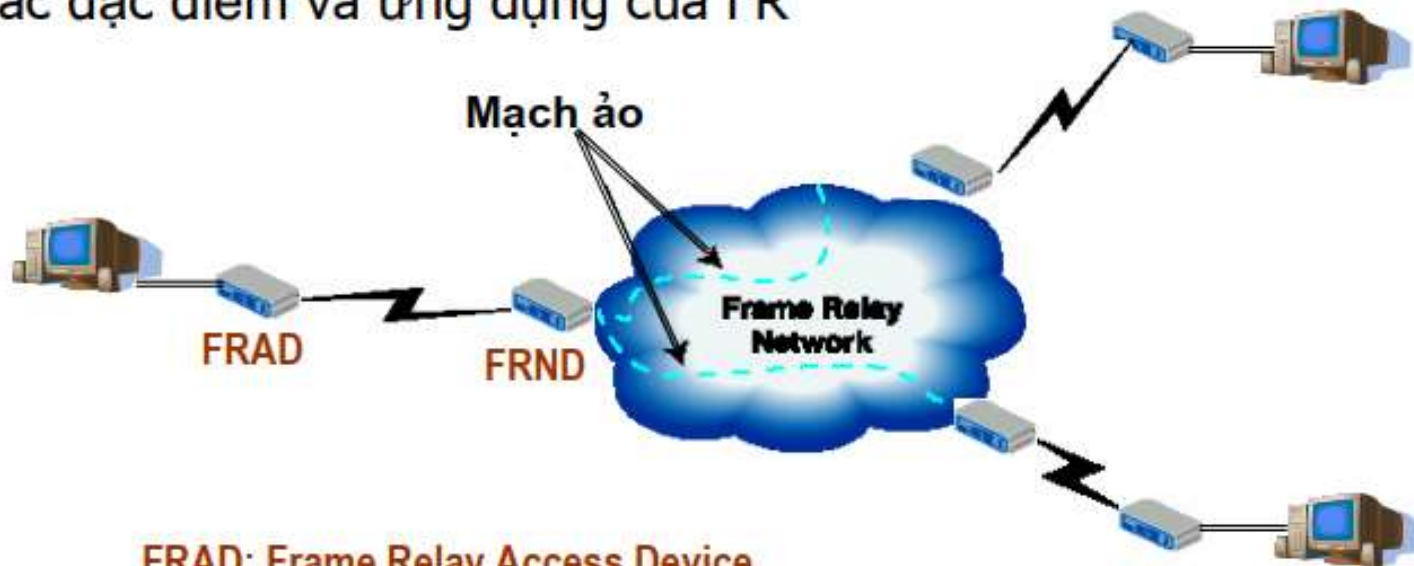


Các công nghệ chuyển gói

- Công nghệ Frame Relay
- Công nghệ ATM
- Công nghệ IP

Công nghệ FR

- Kiến trúc giao thức FR (chuyển tiếp khung)
- Khuôn dạng khung
- Các đặc điểm và ứng dụng của FR



FRAD: Frame Relay Access Device

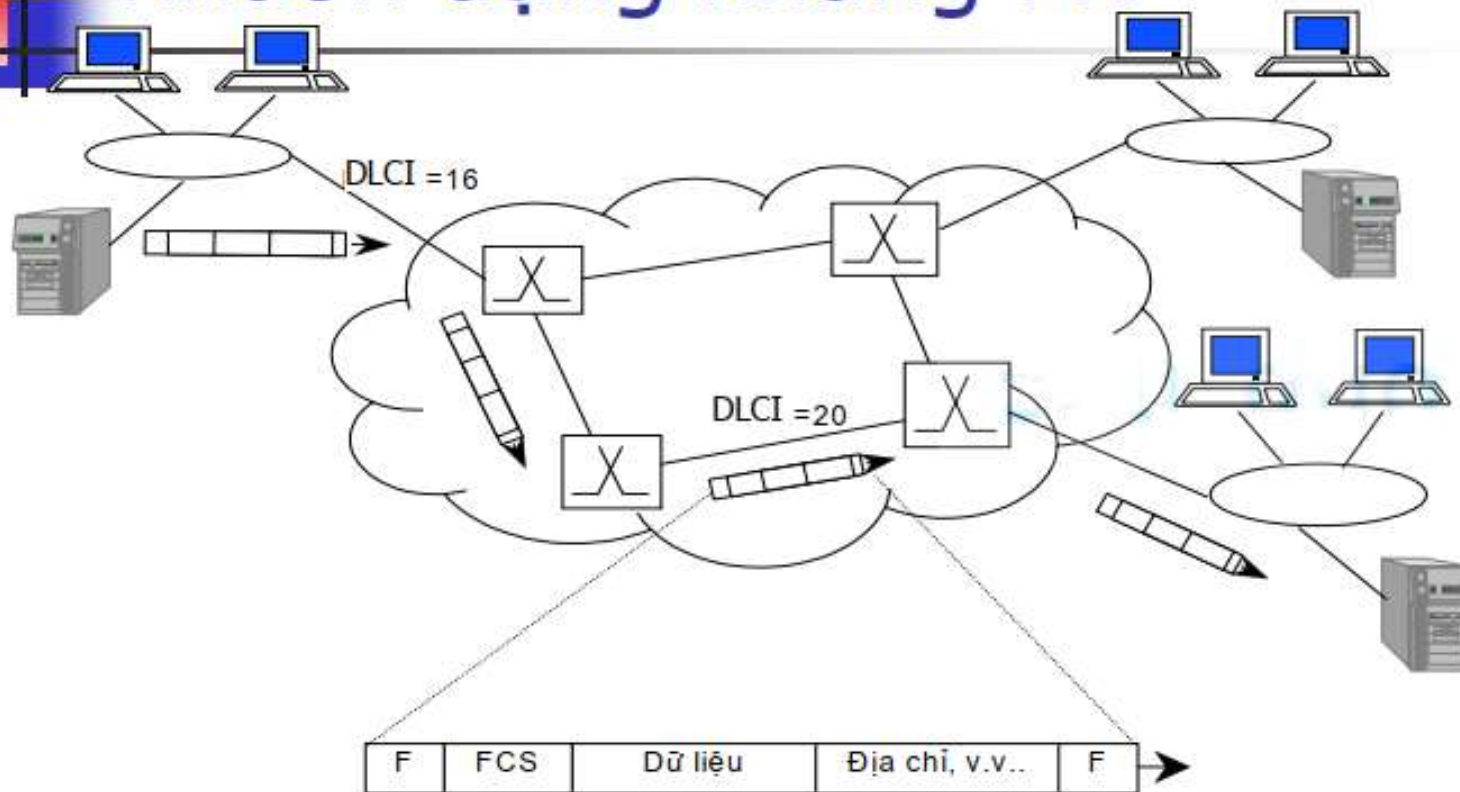
FRND: Frame Relay Network Device



Công nghệ FR

- Kiến trúc giao thức FR
 - Khuyến nghị I.233 và I.122 về phương thức khung, dịch vụ mang (bearer service)
 - Chỉ dùng lớp 1 và lớp 2 đã được đơn giản hóa
 - Lớp 1 cơ bản là lớp vật lý
 - Lớp 2 chỉ sử dụng tiến trình kiểm tra lỗi đơn giản để đảm bảo dữ liệu có lỗi sẽ bị loại bỏ.
 - Các lớp cao hơn ở thiết bị của người sử dụng chịu trách nhiệm cho việc truyền lại dữ liệu lỗi/hỏng.

Khuôn dạng khung FR



- F: Flag
- FCS: Frame Check Sequence
- LCN: Logical Channel Number: Số chỉ kênh logic
- DLCI: Data Link Connection Identifier (nhận dạng kết nối liên kết dữ liệu)



Công nghệ FR

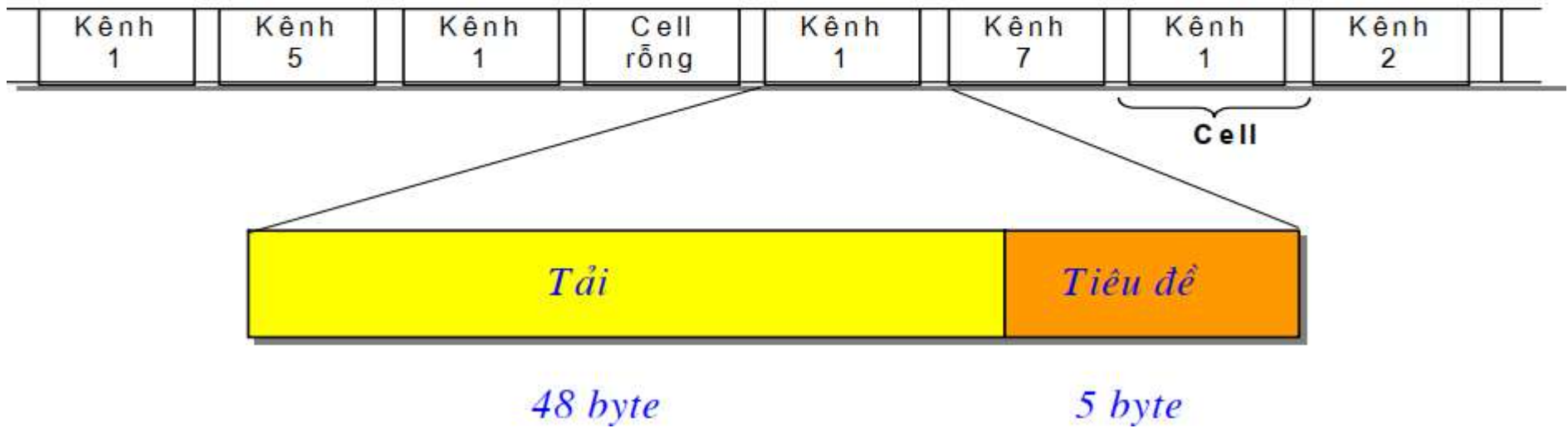
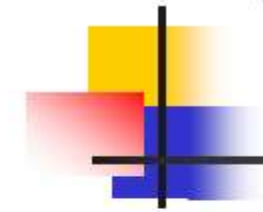
- Các đặc điểm và ứng dụng của FR
 - Kỹ thuật kết nối có liên kết (connection oriented)
 - Độ dài khung (gói) thay đổi
 - Tốc độ cao từ 2-50Mb/s
 - Không cần truyền lại trên từng chặng (chỉ cần ở mức đầu cuối-đầu cuối)
 - Yêu cầu chất lượng truyền dẫn tốt (cáp sợi quang)
 - Chuyển tiếp khung (FR) phần lớn được sử dụng để nối các mạng cục bộ LAN.



Công nghệ ATM

- Kiến trúc giao thức ATM
 - Lớp 1: Lớp vật lý
 - Lớp 2: Lớp ATM
 - Lớp 3: Lớp AAL
- Lớp thích ứng ATM
- Lớp ATM
- Các đặc điểm và ứng dụng của ATM

Công nghệ ATM





Công nghệ ATM

- Lớp thích ứng ATM

- Để cho phép truyền tải các dịch vụ dữ liệu và dịch vụ đẳng thời, thông tin phải được làm thích ứng với mạng trong các cách khác nhau.
 - ATM đã chia thành 4 lớp dịch vụ (A, B, C, D) trên cơ sở của 3 tham số (đồng bộ-tốc độ bit-phương thức truyền tải).
 - Bốn giao thức lớp thích ứng (AAL-1,2,3/4 và 5) được định nghĩa cho mỗi loại
-

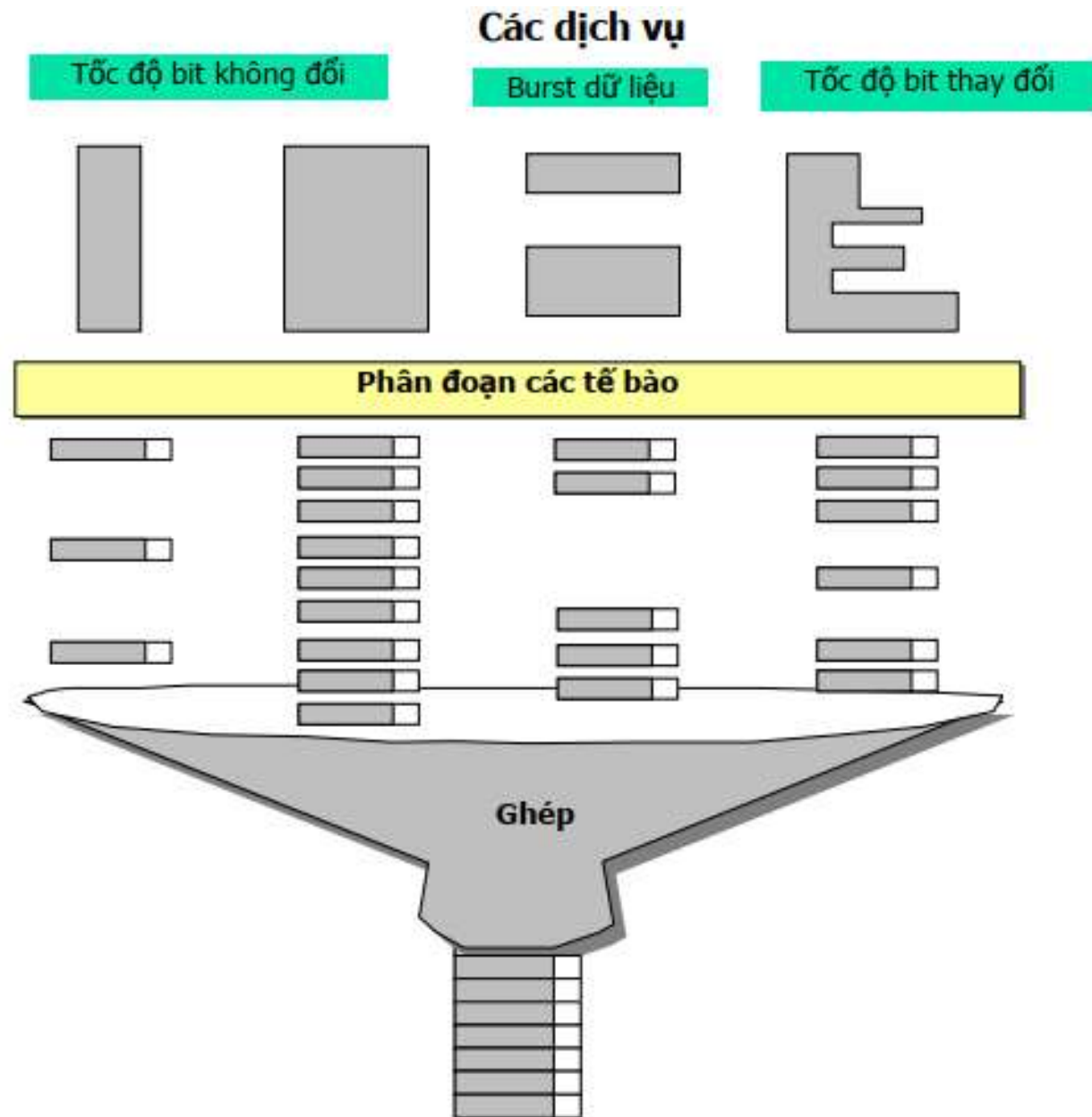
Công nghệ ATM

■ Lớp thích ứng ATM

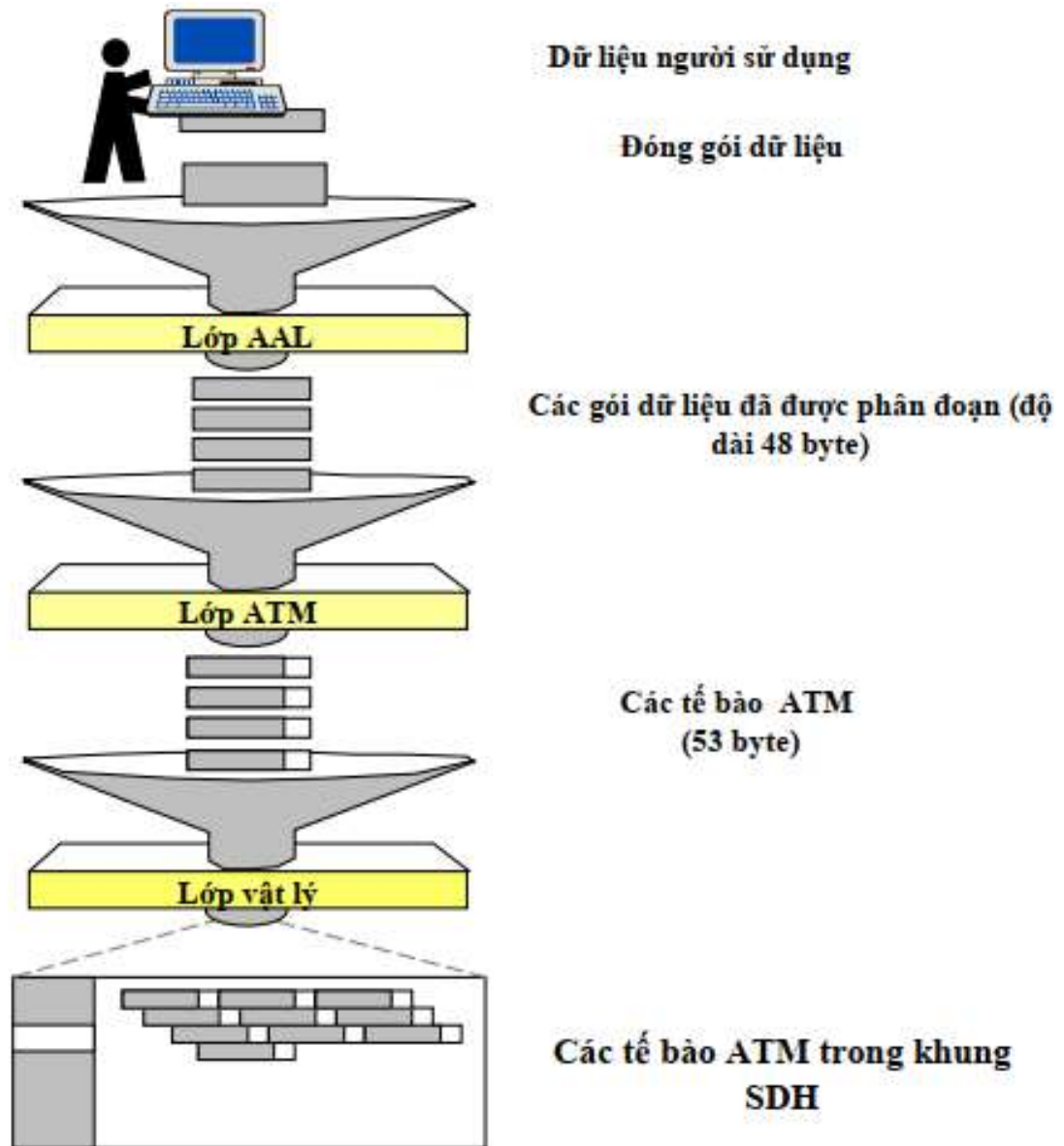
Tiêu đề tế bào 5 octet
AAL-1-5
Các trường thông tin còn lại 44-47 octet

Loại A (Ví dụ: thoại)	Loại B (Ví dụ: Video)	Loại C	Loại D
AAL-1	AAL-2	AAL-3/4 AAL-5	AAL-3/4
Các dịch vụ đồng bộ		Các dịch vụ không đồng bộ	
Tốc độ bit không đổi	Tốc độ bit biến đổi		
Chuyển giao có liên kết			Chuyển giao không liên kết

- Phân đoạn và ghép nối các dịch vụ băng rộng khác nhau



- Lớp Vật lý, lớp ATM và lớp AAL





Công nghệ ATM

- Các đặc điểm và ứng dụng của ATM
 - Kỹ thuật theo kiểu tế bào với độ dài cố định
 - Sử dụng phương thức kết nối định hướng (connection-oriented) cho truyền thông tin
 - Tế bào ATM : gói có độ dài cố định là 53 octet
 - 5 byte tiêu đề (header)
 - 48 byte còn lại dành cho trường thông tin (dữ liệu người sử dụng)
 - Băng tần mềm dẻo
 - Tốc độ rất cao
 - Phù hợp với mọi dịch vụ

4.3. Packet Switching (cont.)

Characteristics:

- ✓ There are two types of packet switching, datagram switching and virtual circuit switching
- ✓ Since packets have a strictly defined maximum length, they can be stored in main memory instead of disk, therefore access delay and cost are minimized.
- ✓ The transmission speeds, between nodes, are optimized. With current technology, packets are generally accepted onto the network **on a first-come, first-served basis**. If the network becomes overloaded, packets are delayed or discarded ("dropped").

4.3. Packet Switching (cont.)

- The size of the packet can vary from 180 bits, the size for the Datakit® virtual circuit switch designed by Bell Labs for communications and business applications
- To 1,024 or 2,048 bits for the 1PSS® switch, also designed by Bell Labs for public data networking
- To 53 bytes for ATM switching, such as Lucent Technologies' packet switches.

4.3. Packet Switching: Datagram

Characteristics:

- Datagram packet switching is like message switching in that each packet is a self-contained unit with complete addressing information attached.
- **None-oriented connection:** each package with the same destination address, do not follow the same route, and they may arrive out of sequence at the exit point node (or the destination).
- **Disadvantage:** take much time for transmission and get more ability for congestion on busy hours.

4.3. Packet Switching: Virtual Circuit

- **Oriented connect:** a preplanned route is established before any data packets are sent.
- A logical connection is established when, a sender send a "call request packet" to the receiver and the receiver send back an acknowledge packet "call accepted packet" to the sender if the receiver agrees on conversational parameters.
- The conversational parameters can be maximum packet sizes, path to be taken, and other variables necessary to establish and maintain the conversation.
- Virtual circuits imply acknowledgements, flow control (sliding window), and error control (Idle RQ, ARQ,..), so virtual circuits are reliable.
- That is, they have the capability to inform upper-protocol layers if a transmission problem occurs.

4.3. Packet Switching: Virtual Circuit (cont.)

- In virtual circuit, the route between stations does not mean that this is a dedicated path, as in circuit switching. It can be rerouted again in case of dropping
- A packet is still buffered at each node and queued for output over a line.
- The difference between virtual circuit and datagram approaches:
 - With virtual circuit, the node does not need to make a routing decision for each packet.
 - It is made only once for all packets using that virtual circuit.
 - Virtual circuit allows to transmit information in continuously, so it is suitable for voice and video information

4.3. Packet Switching: Virtual Circuit (cont.)

- **VC's offer guarantees that,**

The packets sent arrive in the order sent, with no duplicates or omissions, no errors (with high probability) and regardless of how they are implemented internally.

4.3. Packet Switching (cont.)

- **Advantages:**

- ✓ **Cost effective:** because switching devices do not need massive amount of secondary storage.
- ✓ **Offers improved delay characteristics:** because there are no long messages in the queue (maximum packet size is fixed). Packet can be rerouted if there is any problem, such as, busy or disabled links.
- ✓ **Many network users can share the same channel at the same time.**
- ✓ Can maximize link efficiency by making optimal use of link bandwidth.

4.3. Packet Switching (cont.)

- ***Disadvantages:***

- ✓ Protocols for packet switching are typically more complex.
- ✓ It can add some initial costs in implementation.
- ✓ If packet is lost, sender needs to retransmit the data.
- ✓ Another disadvantage is that packet-switched systems still can't deliver the same quality as dedicated circuits in applications requiring very little delay - like voice conversations or moving images.

Problems

1. Draft the general diagram of a wire transmission for analog signal.
2. Draft the general diagram of a wire transmission for digital signal.
3. Draft the general diagram of a wireless transmission for analog signal.
4. Draft the general diagram of a wireless transmission for digital signal.
5. What is skin effect in RF signal transmission? How to overcome the skin effect?
6. What are circuit switching, message switching and packet switching? And what are the suitable forms to use these switching techniques?
7. What types of circuit switching have been using ?
8. What types of package switching have been using?
9. Compare the advantage between circuit switching and package switching.
10. Compare the advantage between datagram switching and virtual circuit switching.
11. Define switching in light of transmission.
12. Define calling rate and holding time.
13. Define lost calls using the terms offered traffic and carried traffic
14. When dealing with traffic formulas and resulting traffic tables, we have to know how lost calls are "handled." What are the three ways that lost calls may be handled?

Exercises

1. Design a SDS of a telephone system. It is required to service for 128 subscribers with blocking probability at 0.02 and inlet utilization probability at 0.15. Calculate the cross points of this system.
2. The statistics during the BH for a particular exchange is 5 lost calls in 841 offered calls. What is the grade of service?
3. Consider one traffic relation that has been designed to meet grade of service objectives. We vary one characteristic—the number of traffic channels. Argue the case for efficiency for 10 circuits versus 49 circuits during the BH. Assume grade of service as 0.01 and \$1.00 per erlang